Understanding Multilateral Tariffs and Exchange Rates*

Jason Lu
International Monetary Fund
Research Department
jlu2@imf.org

Dimitre Milkov
International Monetary Fund
Research Department
dmilkov@imf.org

November 1, 2026

Abstract

The standard prediction that an import tariff appreciates the tariffing country's currency comes from a two-country setting, where the bilateral exchange rate, effective exchange rate, and the terms of trade coincide. In a multilateral setting, these measures separate and respond differently to tariffs. Under symmetric trade relationships, we show that a unilateral tariff appreciates the effective exchange rate but can depreciate the home currency against untariffed bystanders when substitution across foreign suppliers is sufficiently elastic. Tariff retaliation can instead lead both counterparts to depreciate in effective terms. These results are robust to moderate asymmetry, but under strong asymmetry, tariffs on small trading partners can worsen the tariffing country's terms of trade and reduce welfare when foreign suppliers are sufficiently close substitutes. We use the framework to contrast exchange-rate responses during the 2025 tariff episode.

Keywords: multilateral tariffs, exchange rates, trade diversion, terms of trade, Marshall–Lerner condition.
JEL codes: F13, F31, F41.

*Preliminary draft. The views expressed are those of the authors and do not necessarily represent the views of the IMF, its Executive Board, or IMF management.

1 Introduction

What happens to exchange rates when a country imposes a tariff? In the standard two-country model, a tariff shifts expenditure away from imports and improves the trade balance at the prevailing exchange rate. Under the Marshall–Lerner condition, external balance is restored through an appreciation of the tariffing country's currency (Dornbusch, 1974; Krugman, 1982). Because there is only one trading partner, the bilateral exchange rate is also the effective exchange rate. In the long run, in real terms, this same relative price also governs the terms of trade. Thus, in the simplest bilateral setting, the bilateral exchange rate, the effective exchange rate, and the long-run terms of trade all necessarily coincide.

These concepts separate once trade is multilateral. A tariff on one partner changes not only the relative price of foreign and domestic goods, but also relative prices across foreign suppliers. It can therefore affect the exchange rate against the target, the rates against untariffed bystanders, and the effective exchange rate differently. In any differentiable multi-country trade environment, we show that the first-order response can be represented as the adjustment of exchange rates to a vector of tariff-induced trade-balance disturbances. We define a multilateral Marshall–Lerner condition, implied by gross substitutability of trade balances in currency values, that ensures well-behaved adjustment. Crucially, we show that even under this condition, the sign of an individual bilateral exchange-rate response is not determined. A tariff therefore need not appreciate the tariffing country's currency against every partner.

We first answer this question in a general trade environment under symmetry among trading partners. We show that any vector of multilateral tariff changes decomposes into average protection and relative treatment. Average protection uniquely determines the change in the nominal effective exchange rate, while relative treatment determines the relative changes in bilateral exchange rates. Under a unilateral tariff, these two components reinforce each other against the targeted partner but work in opposite directions against untariffed bystanders. The response against bystanders is therefore ambiguous and depends on the structure of the trade environment.

To characterize these counteracting effects, we develop a nested CES Armington model that distinguishes substitution between domestic and foreign goods from substitution across foreign suppliers. Substitution toward domestic goods supports the conventional appreciation, while substitution across foreign suppliers shifts expenditure toward untariffed countries and strengthens their currencies relative to the tariffing country. Allowing these two margins to differ, we show that the tariffing country can depreciate against an untariffed bystander when cross-foreign substitution is sufficiently strong relative to home-foreign substitution, but, under weak home bias, this outcome cannot arise with a single elasticity across domestic and foreign varieties. Empirical estimates distinguishing cross-foreign from home-foreign substitution support treating these margins separately (Broda and Weinstein, 2006; Feenstra et al., 2018; Li et al., 2022).

The same mechanisms extend naturally from unilateral tariffs to retaliation and multilateral trade wars. When the target retaliates with an equal tariff, the average-protection effects on the bilateral exchange rate cancel, while the relative-treatment effects reinforce each other against third

countries. Retaliation therefore strengthens the relative-treatment force and lowers the threshold for depreciation against bystanders from $\rho > N\rho_D$ under a unilateral tariff to $\rho > \rho_D$, where ρ is the elasticity of substitution across foreign suppliers and ρ_D is a domestic diversion margin. In a multilateral trade war, in which every partner retaliates against the tariffing country, the same condition leads it to depreciate against every partner.

Our trade framework is deliberately stylized and is not intended to provide a quantitative account of the 2025 tariff episode, but it nonetheless offers a new perspective on the exchange-rate movements during this period. Our application to 2025 is neither a calibration nor a causal test: it compares the sign patterns the model associates with each phase's tariff configuration with the observed exchange-rate movements. The dollar's depreciation following the tariff announcements was initially interpreted in part as evidence of higher risk premia, declining convenience yields, and concerns about the dollar's global role (Itskhoki and Mukhin, 2025; Kalemli-Özcan et al., 2026). Applying our framework to the broad features of the episode, we show that dollar depreciation need not originate outside the trade channel, particularly when retaliation is expected. Consistent with this possibility, Ostry et al. (2025) document that the dollar tends to appreciate following unilateral US tariffs, depreciate when trading partners retaliate, and remain comparatively stable against the renminbi during US–China tit-for-tat episodes. They hypothesize that dominant-currency pricing may help explain this pattern. Our results show that the same pattern can arise from canonical trade substitution and diversion effects once trade and exchange-rate adjustment is considered in a multilateral setting.

In a general asymmetric setting, where economies differ in size, trade shares, and productivities, exchange-rate responses no longer admit simple analytical expressions that characterize when one force dominates another. Under asymmetry, uniform protection retains the conventional local aggregate terms-of-trade and welfare result, while discriminatory protection can overturn it. In our numerical comparative statics, aggregate reversal arises most naturally when the target is small or lightly weighted, the bystanders account for a large share of trade, and cross-origin substitution is sufficiently strong.

These reversals apply to discriminatory tariffs, not to uniform protection. Under general conditions allowing country sizes, trade shares, home bias, and preference parameters to differ, we prove that a small uniform tariff imposed from a balanced free-trade baseline appreciates the tariffing country's currency weakly against every partner under the multilateral Marshall–Lerner condition. It therefore improves the aggregate terms of trade and raises welfare to first order. This result recovers the classical terms-of-trade argument for uniform protection while showing why its partner-specific counterpart can fail. Uniform protection eliminates relative treatment across suppliers and switches off the cross-origin diversion mechanism that drives the reversals.

Our paper builds on the classical analysis of tariffs under flexible exchange rates, which identifies the conditions under which tariffs lead to exchange-rate appreciation (Lerner, 1936; Metzler, 1949; Mundell, 1961; Dornbusch, 1974; Eichengreen, 1981; Krugman, 1982). A large open-economy macroeconomics literature extends this analysis to environments with nominal rigidities, mone-

tary policy, currency invoicing, and alternative uses of tariff revenue (Lindé and Pescatori, 2019; Barattieri et al., 2021; Erceg et al., 2023; Bergin and Corsetti, 2023; Werning et al., 2025). We instead revisit the flexible-price benchmark in a multilateral setting, where tariffs affect not only the importing country's aggregate exchange rate but also relative exchange rates across trading partners. Our bystander results are related to the literature on third-country effects of preferential trade policy, particularly the roles of trade diversion and competition among exporters (Viner, 1950; Mundell, 1964; Bond and Syropoulos, 1996; Bagwell and Staiger, 1999, 2002). We show how these forces affect bilateral and effective exchange rates when the importing country also produces a competing good. Finally, our multilateral Marshall–Lerner condition draws on the classical literature on stability in multiple markets (Hicks, 1939; Mosak, 1944; Metzler, 1945; Arrow and Hurwicz, 1958), applying these conditions to the exchange-rate response to tariffs in a multilateral setting.

Our results also relate to the empirical literature on the exchange-rate effects of tariffs, including the 2025 episode (Itskhoki and Mukhin, 2025; Kalemli-Özcan et al., 2026; Ostry et al., 2025) and earlier episodes (Furceri et al., 2019; Matveev and Ruge-Murcia, 2021; Jeanne and Son, 2024). Our analysis also offers a complementary perspective on the empirical literature estimating tariff pass-through into import prices (Amiti et al., 2019; Fajgelbaum et al., 2020; Cavallo et al., 2021). This literature typically uses dollar-denominated import prices as a proxy for tariff incidence. In a multilateral setting, however, exchange-rate movements provide an additional margin of incidence through changes in relative prices. An exporter whose dollar price is unchanged may still experience a decline in the home-currency value of its receipts when the dollar depreciates. Assessing tariff incidence from import prices alone can therefore overlook adjustment through exchange rates, a margin that may have been particularly relevant during the broad dollar depreciation in 2025.

This paper is organized as follows. Section 2 develops the general multilateral adjustment framework and decomposes tariff changes into average protection and relative treatment. Section 3 derives the results for unilateral tariffs in a nested-CES model. Section 4 analyzes retaliation, connects the results to preferential trade, and applies the framework to the 2025 tariff episode. Section 5 examines asymmetry. Section 6 concludes. Derivations and proofs are in the appendix, and secondary results are in an online supplement.

2 Tariffs and Exchange-Rate Adjustment

2.1 The bilateral benchmark

Consider two countries, A and B. Let e_{AB} denote units of A's currency per unit of B's, so that a rise is a depreciation of A, and write $\nu \equiv \log e_{AB}$. Country A levies an ad valorem tariff τ on imports from B.

Each country's trade balance TB_i is the value of its exports minus the value of its imports, both at border (pre-tariff) prices and both expressed in one fixed common unit of account. Tariff revenue is a domestic transfer: it is counted neither as export revenue nor as an import payment, although its rebate or other use may affect income, quantities and therefore the trade balance. In

the nested-CES model of Section 3 it is rebated lump-sum. Let v_l denote the common-unit value of one unit of currency l. We select A as numeraire and normalize $v_A = 1$, so these trade-balance values are numerically equal to values expressed in A's currency. This convention is used throughout this section and in the proof of Lemma 2.2, and all derivatives in this section are evaluated at the balanced baseline unless otherwise indicated.

Equilibrium with balanced trade is $\mathrm{TB}_A(\nu, \tau) = 0$. Linearizing,

$$d\,\mathrm{TB}_A = \mathrm{TB}_{A\nu}\,d\nu + \mathrm{TB}_{A\tau}\,d\tau = 0 \quad \Longrightarrow \quad d\nu^* = -\frac{\mathrm{TB}_{A\tau}}{\mathrm{TB}_{A\nu}}\,d\tau, \tag{2.1}$$

where $\mathrm{TB}_{A\nu} \equiv \partial \mathrm{TB}_A/\partial\nu$ holds the tariff fixed and $\mathrm{TB}_{A\tau} \equiv \partial \mathrm{TB}_A/\partial\tau$ holds ν fixed. At a fixed exchange rate, the tariff produces a positive trade-balance disturbance, $\mathrm{TB}_{A\tau} > 0$, when the induced decline in import payments exceeds any decline in export receipts. Conditional on this disturbance, Eq. (2.1) implies an appreciation if and only if a depreciation improves the trade balance, $\mathrm{TB}_{A\nu} > 0$.

To express $\mathrm{TB}_{A\nu}$ in elasticities, let Q_X and Q_M be A's export and import quantities, and hold border prices fixed in the producer's currency, as in the nested-CES model, with units chosen so that each border price is one. A's export receipts are then Q_X and its import payments $e_{AB}Q_M$, both in A's currency, so $\mathrm{TB}_A = Q_X - e^{\nu}Q_M$. Define

$$\varepsilon_X \equiv \frac{\partial \log Q_X}{\partial \nu}, \qquad \varepsilon_M \equiv -\frac{\partial \log Q_M}{\partial \nu},$$

at fixed τ and at the balanced baseline. Both are positive when a depreciation raises export volume and lowers import volume. Let $X \equiv Q_X = e_{AB}Q_M$ be the common baseline border value of exports and imports. Differentiating, $\mathrm{TB}_{A\nu} = Q_X\varepsilon_X - e_{AB}Q_M(1-\varepsilon_M) = X(\varepsilon_X + \varepsilon_M - 1)$: the -1 is the valuation effect of a depreciation on the payment for each imported unit. The condition $\mathrm{TB}_{A\nu} > 0$ is therefore the Marshall–Lerner condition $\varepsilon_X + \varepsilon_M > 1$. Figure 1 illustrates; the parallel shift it shows is a first-order approximation around the baseline.

The bilateral appreciation result therefore requires two conditions: the tariff must improve the trade balance at the initial exchange rate, $\mathrm{TB}_{A\tau} > 0$, and a depreciation must improve the trade balance, $\mathrm{TB}_{A\nu} > 0$. Appendix C.1 states sufficient conditions for both. With three or more countries, however, these conditions no longer sign every bilateral exchange-rate response.

2.2 Multilateral adjustment

Now consider N countries, indexed by $\mathcal{I} = \{1, \ldots, N\}$, and let $\mathcal{I}_{-A} = \mathcal{I}\setminus\{A\}$. Take A's currency as numeraire. The $N-1$ bilateral rates e_{Aj}, $j \in \mathcal{I}_{-A}$, span all exchange-rate changes. Trade balances sum to zero (Walras' law), which leaves $N-1$ independent trade-balance conditions, indexed by $i \in \mathcal{I}_{-A}$:

$$\mathrm{TB}_i\big(\{e_{Aj}\}_{j\in\mathcal{I}_{-A}}, \{\tau_{jk}\}_{(j,k)\in\mathcal{T}}\big) = 0, \qquad i \in \mathcal{I}_{-A},$$

where τ_{jk} is country j's ad valorem tariff on imports from k and $\mathcal{T} \equiv \{(j,k) \in \mathcal{I}\times\mathcal{I} : j \neq k\}$ is the set of all $N(N-1)$ ordered country pairs. Stack the bilateral rates as $\nu = (\nu_l)_{l\in\mathcal{I}_{-A}}$ with

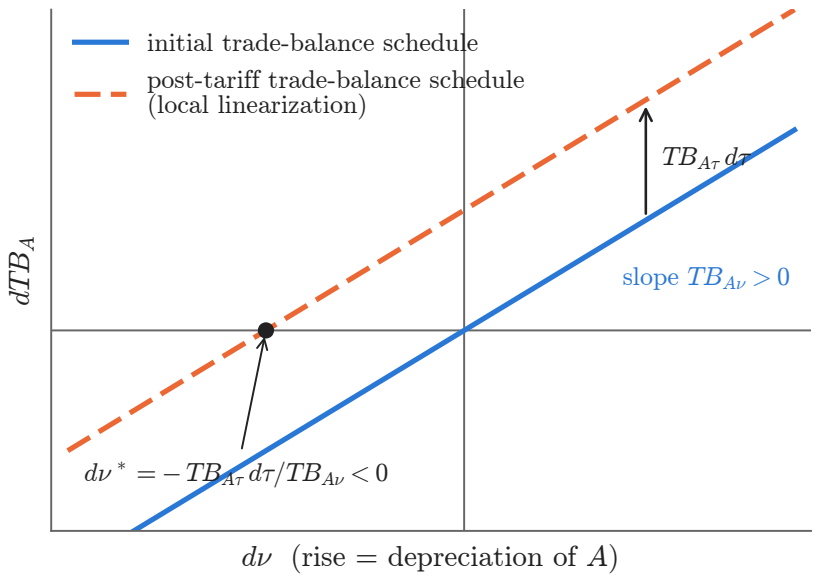


Figure 1: Bilateral tariff adjustment. At a fixed exchange rate, the tariff improves the trade balance; under Marshall–Lerner, equilibrium is restored through appreciation.

$\nu_l = \log e_{Al}$, in increasing order of l, and the tariffs as $\tau = (\tau_{jk})_{(j,k)\in\mathcal{T}}$, in lexicographic order of (j,k). Linearizing,

$$J\, d\nu + G\, d\tau = 0, \qquad J_{il} \equiv \left.\frac{\partial \mathrm{TB}_i}{\partial \nu_l}\right|_{\tau}, \qquad G_{i,(jk)} \equiv \left.\frac{\partial \mathrm{TB}_i}{\partial \tau_{jk}}\right|_{\nu}, \tag{2.2}$$

for $i, l \in \mathcal{I}_{-A}$ and $(j,k) \in \mathcal{T}$, so $J \in \mathbb{R}^{(N-1)\times(N-1)}$ and $G \in \mathbb{R}^{(N-1)\times N(N-1)}$. Consider a tariff change along a fixed direction,

$$d\tau = a\, ds, \qquad a \in \mathbb{R}^{N(N-1)}, \quad \|a\|_2 = 1,$$

where the unit norm serves only to normalize the scalar s. Define $F \equiv Ga$, the marginal vector of trade-balance disturbances per unit increase in s, evaluated at unchanged exchange rates. Substituting into Eq. (2.2),

$$J\,\frac{d\nu}{ds} + Ga = 0.$$

Provided J is nonsingular, therefore,

$$\boxed{\frac{d\nu}{ds} = (-J)^{-1}F.} \tag{2.3}$$

Equation (2.3) separates the trade-balance effects of the tariff at unchanged exchange rates, F, which include expenditure switching, tariff revenue and income effects, from the exchange-rate movements required to restore external balance, $(-J)^{-1}$. In the bilateral case both are scalars, and Eq. (2.3) reduces to Eq. (2.1). With $N \geq 3$, however, the elements of F need not have the

same sign: a tariff imposed by A on B can improve a bystander C's trade balance at unchanged exchange rates. The response of any bilateral exchange rate, including ν_C, therefore depends on the full vector of tariff-induced trade-balance disturbances, not on its own element alone.

2.3 A multilateral Marshall–Lerner condition

Definition 2.1 (Multilateral Marshall–Lerner condition, MML)**.** Let $i, l \in \mathcal{I}_{-A}$. The adjustment mechanism satisfies MML if $J_{il} \geq 0$ for every $l \neq i$, and $(-J)^{-1}$ exists with $(-J)^{-1} \geq 0$ componentwise, that is, $[(-J)^{-1}]_{il} \geq 0$ for all i, l. Equivalently, $-J$ is a nonsingular M-matrix.

Under MML, writing $\vartheta_{ij} \equiv [(-J)^{-1}]_{ij} \geq 0$,

$$\frac{d\nu_i}{ds} = \sum_{j \in \mathcal{I}_{-A}} \vartheta_{ij} F_j, \qquad i \in \mathcal{I}_{-A}. \tag{2.4}$$

Every disturbance enters every bilateral response with a nonnegative weight. Hence, componentwise, if $F \geq 0$ then $d\nu/ds \geq 0$, and if $F \leq 0$ then $d\nu/ds \leq 0$, where a positive $d\nu_i/ds$ is a depreciation of A against i. This is the multilateral counterpart of the two-country result.

A primitive sufficient condition for MML is gross substitutability of trade balances, which is stated in terms of the full $N \times N$ Jacobian with respect to currency values. Let $\chi_l \equiv \log v_l$, $l \in \mathcal{I}$, be the log common-unit value of currency l. Since $e_{Aj} = v_j / v_A$,

$$\nu_j = \chi_j - \chi_A, \qquad j \in \mathcal{I}_{-A}.$$

Define $\widetilde{J}_{il} \equiv \partial \mathrm{TB}_i / \partial \chi_l$ for $i, l \in \mathcal{I}$, at fixed tariffs. If each trade balance is homogeneous of degree one in currency values, Euler's identity gives $\sum_{l \in \mathcal{I}} \widetilde{J}_{il} = \mathrm{TB}_i = 0$ at the balanced baseline, and Walras' law, $\sum_{i \in \mathcal{I}} \mathrm{TB}_i \equiv 0$, gives $\sum_{i \in \mathcal{I}} \widetilde{J}_{il} = 0$ for every $l \in \mathcal{I}$. Because χ_A is fixed by the normalization $v_A = 1$, a change in ν_l is a change in χ_l alone, so the reduced Jacobian of Eq. (2.2) is $J = (\widetilde{J}_{il})_{i,l \in \mathcal{I}_{-A}}$. Trade balances are gross substitutes if

$$\widetilde{J}_{il} \geq 0 \qquad \text{for all } i, l \in \mathcal{I},\ l \neq i,$$

that is, if an appreciation of currency l, a rise in χ_l, weakly improves every other country's trade balance.

Lemma 2.2. *If each trade balance is homogeneous of degree one in currency values, $\widetilde{J}_{il} \geq 0$ for all $i, l \in \mathcal{I}$ with $l \neq i$, the reduced matrix $J = (\widetilde{J}_{il})_{i,l \in \mathcal{I}_{-A}}$ is irreducible, and $\widetilde{J}_{iA} > 0$ for at least one $i \in \mathcal{I}_{-A}$, then MML holds.*

Lemma 2.2 is the trade-balance analogue of the gross-substitutes stability results of Hicks (1939), Mosak (1944), Metzler (1945) and Arrow et al. (1959). MML ensures that multilateral adjustment is well behaved, and a disturbance vector of one sign produces exchange-rate responses of the same sign. When F contains both positive and negative elements, however, MML alone does not determine the sign of an individual bilateral exchange-rate response.

2.4 Symmetric trading partners

Equation (2.3) holds at any balanced baseline. When the tariffing country's partners are symmetric, it can be solved using two eigenspace-specific eigenvalues of $-J$ and two of $G^{(A)}$.

Restrict attention to tariff changes imposed by A, $t = (t_j)_{j\in\mathcal{I}_{-A}}$ with $t_j \equiv d\tau_{Aj}$, and let $G^{(A)} \in \mathbb{R}^{(N-1)\times(N-1)}$ be the corresponding restriction of the tariff-incidence matrix,

$$G^{(A)}_{ij} \equiv \left.\frac{\partial \mathrm{TB}_i}{\partial \tau_{Aj}}\right|_{\nu}, \qquad i,j \in \mathcal{I}_{-A},$$

so that $J\,d\nu + G^{(A)}t = 0$. Assume that the primitives and the balanced baseline allocation are invariant to every relabeling of A's $N-1$ partners. This implies that, for every $(N-1)\times(N-1)$ partner-permutation matrix P,

$$PJP' = J \qquad \text{and} \qquad PG^{(A)}P' = G^{(A)}.$$

A free-trade or common-external-tariff baseline is an example only when the partner primitives are also symmetric. We then call A's partners symmetric. With $\mathbf{1}$ the $(N-1)$-vector of ones, decompose

$$t = \bar{t}\,\mathbf{1} + \widetilde{t}, \qquad \bar{t} = \tfrac{1}{N-1}\mathbf{1}'t, \qquad \mathbf{1}'\widetilde{t} = 0. \tag{2.5}$$

We call $\bar{t}$ *average protection* and $\widetilde{t}$ *relative treatment*. Any permutation-invariant matrix has the form $aI + b\mathbf{1}\mathbf{1}'$, so the one-dimensional common subspace $\mathrm{span}\{\mathbf{1}\}$ and the $(N-2)$-dimensional deviation subspace $\{v : \mathbf{1}'v = 0\}$ are eigenspaces of both $G^{(A)}$ and $-J$:

$$G^{(A)}\mathbf{1} = g_S\mathbf{1}, \quad G^{(A)}v = g_D v; \qquad (-J)\mathbf{1} = j_S\mathbf{1}, \quad (-J)v = j_D v \qquad (\mathbf{1}'v = 0).$$

The eigenvalues j_D and g_D each have multiplicity $N-2$.

We measure A's effective exchange rate by its nominal effective exchange rate (NEER) with baseline bilateral trade weights. With f_{ij} the border value of country i's imports from j,

$$w_{Aj} \equiv \frac{f_{Aj} + f_{jA}}{\sum_{k\neq A}(f_{Ak} + f_{kA})}, \qquad \sum_{j\neq A} w_{Aj} = 1,$$
$$E_A \equiv \prod_{j\neq A} e_{Aj}^{w_{Aj}}, \qquad d\nu^E_A \equiv d\log E_A = \sum_{j\neq A} w_{Aj}\,d\nu_j,$$

holding the baseline weights fixed for the local comparative static. This is the effective exchange rate throughout the paper. Under partner symmetry the bilateral trade weights are equal, $w_{Aj} = 1/(N-1)$, so in this section the NEER reduces to the simple average of the bilateral log exchange rates, $d\nu^E_A = \frac{1}{N-1}\sum_{j\neq A} d\nu_j$. For partners $j,k \in \mathcal{I}_{-A}$ the cross rate is $e_{jk} = e_{Ak}/e_{Aj}$, units of j's currency per unit of k's.

Proposition 2.3 (Average protection and relative treatment)**.** *Let partners be symmetric, $N \geq 3$,*

and $j_S, j_D \neq 0$, so that J is nonsingular. For any tariff vector t,

$$d\nu = \lambda_{\text{avg}}\,\bar{t}\,\mathbf{1} + \lambda_{\text{rel}}\,\widetilde{t}, \qquad \lambda_{\text{avg}} = \frac{g_S}{j_S}, \quad \lambda_{\text{rel}} = \frac{g_D}{j_D}. \tag{2.6}$$

Hence

$$d\nu_A^E = \lambda_{\text{avg}}\bar{t}, \qquad d\nu_j - d\nu_A^E = \lambda_{\text{rel}}\widetilde{t}_j, \qquad d\log e_{jk} = d\nu_k - d\nu_j = \lambda_{\text{rel}}(t_k - t_j). \tag{2.7}$$

Moreover, under the permutation-symmetry assumptions above, MML is equivalent to $j_S > 0$ and $j_D \geq j_S$.

The two components lie in complementary subspaces, $\mathbb{R}^{N-1} = \text{span}\{\mathbf{1}\} \oplus \{v : \mathbf{1}'v = 0\}$, and each is scaled by its own coefficient. With $N = 3$ these subspaces span a plane, which Fig. 2 uses to illustrate the decomposition.

Proposition 2.3 thus reduces the exchange-rate response to any tariff vector to two coefficients: λ_{avg}, through which average protection moves the common component of A's bilateral rates and hence its effective exchange rate, and λ_{rel}, through which relative treatment moves individual bilateral rates around that component and the cross rates among partners. The decomposition follows from partner symmetry alone and does not rely on the nested-CES structure or any other particular demand system. What remains undetermined are the signs and magnitudes of λ_{avg} and λ_{rel}, and hence the relative strength of the two components. Section 3 introduces a nested-CES model in which these coefficients depend on substitution between domestic and foreign goods and on substitution across foreign suppliers.

3 The Nested-CES Model

This section specializes the decomposition of Section 2.4 into average protection and relative treatment to a nested-CES trade model. In the model, η governs substitution between the domestic tradable and the import composite, and ρ governs substitution across foreign origins. We work locally, to first order around a symmetric, balanced, free-trade benchmark. The central result is that a unilateral tariff always appreciates the tariffing country's NEER and its currency against the target, but depreciates it against untariffed bystanders only when $\rho > N\rho_D$, where ρ_D is a domestic diversion margin defined below. With weak home bias, $\alpha_D \geq 1/N$, a specification with a single elasticity, $\rho = \eta$, cannot produce this reversal.

3.1 Production, preferences and equilibrium

There are $N \geq 2$ countries, $i \in \{1, \ldots, N\}$. Country A is the tariffing country, B is the targeted partner, and lowercase c denotes an individual untariffed bystander; uppercase C is reserved for the composite bystander of Section 3.5. Each country is endowed with L_i units of labor, mobile across sectors, and produces a country-specific tradable and a non-tradable with labor only, $Y_{Ti} = A_{Ti}L_{Ti}$

and $Y_{Ni} = A_{Ni}L_{Ni}$, with $L_{Ti} + L_{Ni} = L_i$. Normalizing each tradable's producer price to one in its own currency gives $w_i = A_{Ti}$ and $P_{Ni} = A_{Ti}/A_{Ni}$. Relative international producer prices therefore move only through exchange rates, and the bilateral exchange rate equals the bilateral terms of trade and the relative wage in a common currency.

Preferences are

$$C_i = C_{Ti}^{\alpha_T} C_{Ni}^{1-\alpha_T}, \quad C_{Ti} = \left[\alpha_D^{1/\eta} C_{Tii}^{\frac{\eta-1}{\eta}} + (1-\alpha_D)^{1/\eta} M_i^{\frac{\eta-1}{\eta}}\right]^{\frac{\eta}{\eta-1}}, \quad M_i = \left[\textstyle\sum_{j\neq i} \beta_{ij}^{1/\rho} C_{Tji}^{\frac{\rho-1}{\rho}}\right]^{\frac{\rho}{\rho-1}}, \tag{3.1}$$

with $0 < \alpha_D < 1$, $0 < \alpha_T \leq 1$, $\eta, \rho > 0$ (read as Cobb–Douglas at unit elasticity), $\beta_{ij} > 0$ and $\sum_{j\neq i} \beta_{ij} = 1$. Country i levies an ad valorem tariff τ_{ij} on imports from j, so the consumer price is $p_{ij} = (1+\tau_{ij})e_{ij}$, and the import and tradable price indices are $P_{Mi} = [\sum_{j\neq i} \beta_{ij} p_{ij}^{1-\rho}]^{1/(1-\rho)}$ and $P_{Ti} = [\alpha_D + (1-\alpha_D)P_{Mi}^{1-\eta}]^{1/(1-\eta)}$. Income is

$$I_i = \frac{w_i L_i}{1 - \sum_{j\neq i} \frac{\tau_{ij}}{1+\tau_{ij}} s_{ij}}, \tag{3.2}$$

where s_{ij} is the tariff-inclusive expenditure share of income on j's good. Tariff revenue is rebated lump-sum and is a domestic transfer, so the external-balance conditions use flows at border prices:

$$\mathrm{TB}_i = \sum_{k\neq i} f_{ki} - \sum_{j\neq i} f_{ij}, \qquad f_{ij} = \frac{s_{ij}\tilde{I}_i}{1+\tau_{ij}}, \tag{3.3}$$

with $\tilde{I}_i = e_{Ai} I_i$ income in A's currency. Under the normalization $v_A = 1$ of Section 2.1, values in A's currency equal values in the common unit of account, so Eq. (3.3) is the trade-balance measure of Section 2. With A's currency as numeraire, the $N-1$ independent conditions $\mathrm{TB}_i = 0$, $i \neq A$, determine the $N-1$ bilateral rates; wages, prices and expenditure shares follow from the technologies and the demand system Eq. (3.1).

We write $\nu_j = \log e_{Aj}$, with $\nu_A = 0$. The NEER is that of Section 2.4, $d\nu_A^E = \sum_{j\neq A} w_{Aj}\, d\nu_j$ with baseline trade weights $w_{Aj} = (f_{Aj} + f_{jA})/\sum_{k\neq A}(f_{Ak} + f_{kA})$. The real exchange rate is $q_{Aj} = e_{Aj}P_j/P_A$, with P_i the consumer price index in own currency. A fall in ν_j is an appreciation of A against j and a rise a depreciation; a rise in $\log q_{Aj}$ is a real depreciation of A against j.

The symmetric benchmark is free trade with $\beta_{ij} = 1/(N-1)$, common parameters, and equal endowments and productivities. It retains home bias α_D. Bilateral trade weights then equal $1/(N-1)$, and $m \equiv \alpha_T(1-\alpha_D)$ denotes the import share of income.

3.2 The two-country benchmark

With $N = 2$ there is a single foreign origin, so ρ is inactive. Linearizing gives $d\mathrm{TB}_A = mM_2\, d\nu_B + m\rho_D\, d\tau$, so

$$\left.\frac{d\log e_{AB}}{d\tau}\right|_{\tau=0} = -\frac{\rho_D}{M_2},$$
$$\rho_D \equiv \underbrace{\alpha_D\eta}_{\text{domestic substitution}} + \underbrace{(1-\alpha_D)(1-\alpha_T)}_{\text{rebate / non-tradables}}, \qquad M_2 \equiv 1 + 2\alpha_D(\eta-1) = \varepsilon_X + \varepsilon_M - 1. \tag{3.4}$$

We call ρ_D the domestic diversion margin. A tariff shifts expenditure toward the domestic economy in two ways: through substitution toward the domestic tradable, and through the income effect of the rebated revenue, part of which is spent on non-tradables. An exchange-rate movement raises no revenue and so has no rebate term, which is why ρ_D differs from M_2. With more than two countries, adjustment also separates into a common movement against all foreign suppliers and relative movements across them.

3.3 The N-country case

The following theorem maps the model's primitives into the coefficients of Proposition 2.3.

Theorem 3.1 (Exchange-rate responses in the symmetric nested-CES model)**.** *At the symmetric benchmark define*

$$M_N \equiv \frac{1}{N-1}\Big[1 + \underbrace{N\alpha_D(\eta-1)}_{\textit{domestic–import adjustment}} + \underbrace{(N-2)\rho}_{\textit{foreign-origin adjustment}}\Big]. \tag{3.5}$$

The eigenvalues of Proposition 2.3 are

$$g_S = -\frac{m\rho_D}{N-1}, \qquad g_D = -\frac{m\rho}{N-1}, \qquad j_S = \frac{mM_N}{N-1}, \qquad j_D = Nj_S. \tag{3.6}$$

MML holds if and only if $M_N > 0$, and $\eta \geq 1$ suffices. When it holds,

$$\lambda_{\mathrm{avg}} = -\frac{\rho_D}{M_N}, \qquad \lambda_{\mathrm{rel}} = -\frac{\rho}{NM_N}, \qquad d\nu_j = \lambda_{\mathrm{avg}}\bar{t} + \lambda_{\mathrm{rel}}\widetilde{t}_j. \tag{3.7}$$

The eigenvalues are derived in Lemma A.1 and Eq. (A.4), and the proof is in Appendix B. M_N is affine in ρ; we write $M_N \equiv M_N(\rho)$ and suppress the argument except in Appendix C.2, where the dependence on ρ matters.

M_N is the sensitivity of A's aggregate trade balance to its effective exchange rate: $d\mathrm{TB}_A = mM_N\, d\nu_A^E$ at the equal weights of the symmetric benchmark. For $N = 2$ it reduces to M_2. For $N > 2$ it includes ρ, because foreign buyers can switch between A's goods and those of other foreign suppliers.

The equality $j_D = Nj_S$ means that zero-sum relative exchange-rate movements are N times as

effective as common movements in closing trade imbalances, which is why λ_{rel} carries the factor $1/N$. A common movement shifts only partners' trade with A, whereas a relative movement also shifts partners' trade with *each other*, and at the symmetric benchmark A accounts for a share $1/N$ of the trade on which relative movements act. Because relative treatment is zero-sum, to first order at the symmetric benchmark it leaves tariff revenue, income and the tradable/non-tradable split unchanged, so its disturbance is governed by ρ alone.

3.4 A unilateral tariff

Let A raise its tariff on B alone, $t = d\tau\, e_B$. Its mean component is $\bar{t} = d\tau/(N-1)$, B's relative component is $\widetilde{t}_B = \frac{N-2}{N-1} d\tau$, and each bystander's relative component is $\widetilde{t}_c = -\frac{d\tau}{N-1}$.

Proposition 3.2 (Unilateral tariff)**.** *At the symmetric benchmark under MML, to first order,*

$$\frac{d\nu_A^E}{d\tau} = -\frac{\rho_D}{(N-1)M_N} < 0, \qquad \frac{d\nu_B}{d\tau} = -\frac{N\rho_D + (N-2)\rho}{N(N-1)M_N} < 0, \qquad \frac{d\nu_c}{d\tau} = \frac{\rho - N\rho_D}{N(N-1)M_N} \tag{3.8}$$

for every untariffed bystander c. Hence:

(i) A's NEER appreciates for every N and ρ.

(ii) A appreciates against the target.

(iii) A depreciates against each untariffed bystander if and only if $\rho > N\rho_D$.

(iv) The same threshold applies when A tariffs any $k \leq N-2$ partners. More generally, for a tariff vector with $\bar{t} > 0$, A depreciates against a partner j receiving a below-average tariff increase, $t_j < \bar{t}$, if and only if $t_j/\bar{t} < 1 - N\rho_D/\rho$, which requires $\rho > N\rho_D$.

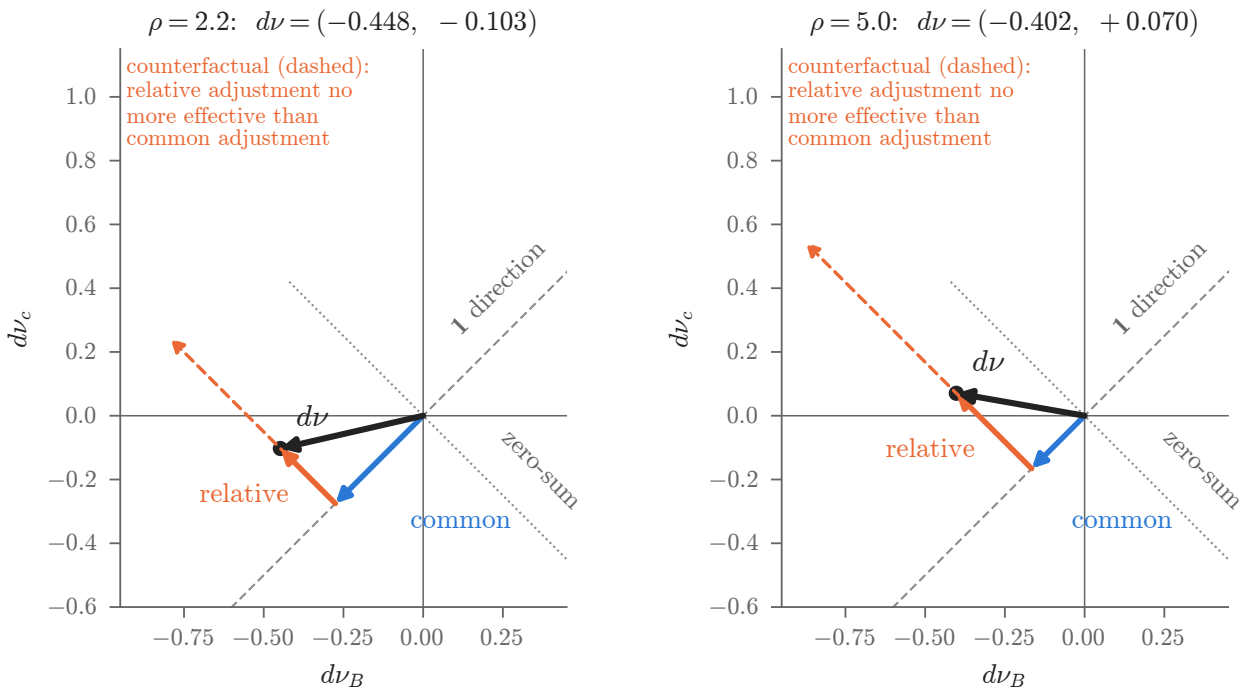
Against the target, average protection and relative treatment push in the same direction. Against a bystander they are offsetting pressures: average protection pushes toward an appreciation of A, while the bystander's favorable relative treatment pushes toward a depreciation.

Corollary 3.3 (No reversal with a single elasticity under weak home bias)**.** *Suppose weak home bias, $\alpha_D \geq 1/N$. Then*

$$N\rho_D - \eta = \eta(N\alpha_D - 1) + N(1-\alpha_D)(1-\alpha_T) \geq 0, \tag{3.9}$$

so $\rho > \eta$ is necessary for a bystander depreciation. In any specification with a single elasticity across all tradable varieties ($\rho = \eta = \sigma$), $\rho - N\rho_D = -[\sigma(N\alpha_D - 1) + N(1-\alpha_D)(1-\alpha_T)] \leq 0$, so wherever MML holds $d\nu_c/d\tau \leq 0$: A never depreciates against a bystander. The response is strictly negative except at the knife edge $\alpha_D = 1/N$, $\alpha_T = 1$, where it is zero for every $\sigma > 0$. More generally, at $\alpha_D = 1/N$ the response is $-(1-\alpha_T)/(N\sigma)$: if $\alpha_T < 1$ it approaches zero from below as $\sigma \to \infty$, and if $\alpha_T = 1$ it is identically zero.

Under weak home bias, the reversal therefore requires cross-origin substitution to exceed home–import substitution. Evidence in Feenstra et al. (2018) and Li et al. (2022) supports allowing ρ



(a) $\rho = 2.2$: $\rho_D < \rho < 3\rho_D$ **(b)** $\rho = 5$: $\rho > 3\rho_D$

Figure 2: Average protection and relative treatment. Unilateral tariff with $N = 3$: (a) $\rho_D < \rho < 3\rho_D$; (b) $\rho > 3\rho_D$. *Notes:* $\alpha_D = 0.7$, $\alpha_T = 0.6$, $\eta = 1.5$.

and η to differ. At an illustrative US-like calibration ($\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$, so $\rho_D = 1.32$), the threshold is $N\rho_D = 3.96$ for $N = 3$ and 6.60 for $N = 5$.

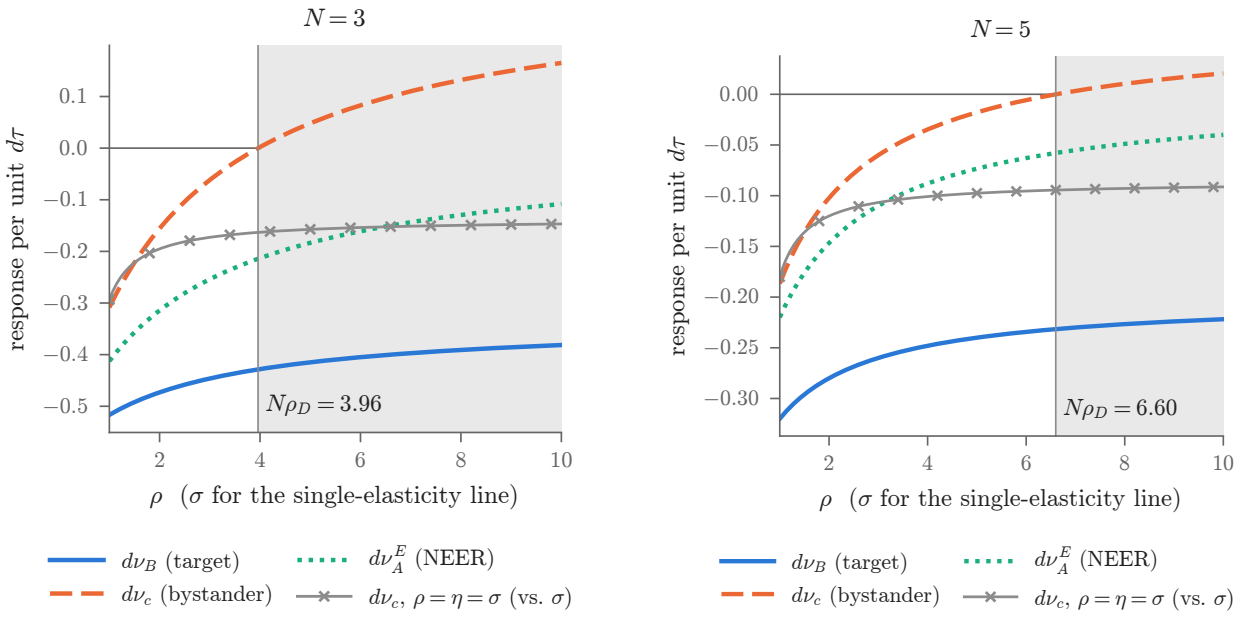
At unchanged exchange rates, the trade balance of an individual bystander improves once $\rho > \rho_D$:

$$F_c = \frac{m(\rho - \rho_D)}{(N-1)^2}. \tag{3.10}$$

This is not sufficient for A to depreciate against the bystander in equilibrium. Under MML the exchange-rate response depends on the full vector of trade-balance disturbances, including the target's negative disturbance, and reverses only under the stronger condition $\rho > N\rho_D$. Thus, for $\rho_D < \rho < N\rho_D$, the bystander gains at the initial exchange rates even though A ultimately appreciates against it (Fig. 2a). The gap between the two thresholds reflects the greater effectiveness of relative exchange-rate adjustment identified in Theorem 3.1: if relative and common movements were equally effective, the bystander's equilibrium response would instead change sign when its own disturbance does. Figure 3 plots the three responses against ρ.

3.5 The role of the bystander bloc

The bystander threshold $N\rho_D$ rises with the number of countries. To see why, note that when the bystanders are symmetric and a tariff shock treats them alike, no relative price within the bystander bloc moves, so the bloc can be collapsed into a single region.



(a) $N = 3$ **(b)** $N = 5$

Figure 3: Exchange-rate responses to a unilateral tariff. The bystander response changes sign at $\rho = N\rho_D$. *Notes:* US-like calibration, $\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$.

Proposition 3.4 (Three-region representation)**.** *At the symmetric benchmark with $N \geq 3$, consider a tariff shock that treats the $N-2$ bystanders symmetrically and leaves tariffs among them unchanged, and suppose $M_N \neq 0$ (MML is sufficient). To first order, the response of ν_B, the common response of each bystander, and A's NEER are exactly reproduced by a three-region economy consisting of A, B and a composite bystander C with income $N-2$, home share of tradables $h_C = \alpha_D + (1-\alpha_D)\frac{N-3}{N-1}$, and home–import elasticity*

$$\boxed{\eta_C = \frac{(N-3)\rho + 2\alpha_D\,\eta}{N-3+2\alpha_D}.} \tag{3.11}$$

The complete mapping of trade weights and tariff shocks is given in Appendix B.

The representation shows that the increase in the bystander threshold with N is not a mechanical consequence of combining more countries into a larger bloc. As the number of bystanders rises, a larger share of the composite bloc's activity consists of trade among the original bystanders. That trade adjusts at the cross-origin elasticity ρ, which changes the bloc's internal adjustment and makes relative exchange-rate movements more effective.

This internal trade structure explains why the unilateral threshold rises to $N\rho_D$. The exact representation is given in Appendix B; a counterfactual that holds the composite bloc's internal elasticity fixed is reported in the online supplement.

A nominal bystander reversal need not imply a real reversal because the tariff also raises A's consumer price index. At the US-like calibration with $N = 3$, the nominal threshold is $3\rho_D = 3.96$,

while the real threshold is 4.93. Appendix C.2 derives the real-exchange-rate results and the conditions under which the real effective exchange rate continues to appreciate.

The nested-CES model therefore gives economic content to the decomposition in Section 2. Average protection operates through the domestic diversion margin ρ_D, while relative treatment operates through cross-origin substitution ρ. Under a unilateral tariff, the two forces reinforce each other against the target but oppose each other against untariffed bystanders. A bystander depreciation requires $\rho > N\rho_D$, while the effective exchange rate continues to appreciate. We next examine how retaliation changes this balance.

4 Retaliation and Multilateral Tariff Configurations

How does retaliation change the balance between average protection and relative treatment? Under a unilateral tariff, average protection pushes the tariffing country's currency toward appreciation, while favorable relative treatment pushes untariffed bystanders in the opposite direction. Retaliation strengthens this second force. As a result, the bystander-depreciation threshold falls from $\rho > N\rho_D$ under a unilateral tariff to $\rho > \rho_D$.

4.1 Retaliation

Proposition 4.1 (Retaliation and trade wars)**.** *At the symmetric benchmark with $N \geq 3$ under MML, to first order:*

(i) Under bilateral retaliation (A tariffs B and B tariffs A by $d\tau$), $d\nu_B = 0$ and

$$\frac{d\nu_c}{d\tau} = \frac{\rho - \rho_D}{(N-1)M_N}, \qquad \frac{d\nu_A^E}{d\tau} = \frac{N-2}{N-1} \cdot \frac{\rho - \rho_D}{(N-1)M_N}. \tag{4.1}$$

Both belligerents depreciate against every bystander, and in effective terms, if and only if $\rho > \rho_D$. Cross rates among bystanders are unchanged.

(ii) Under a multilateral trade war (A tariffs every partner, and every partner retaliates against A), $d\nu_j = d\nu_A^E = \frac{(N-2)(\rho-\rho_D)}{(N-1)M_N}\, d\tau$ for all j, so A depreciates against every partner if and only if $\rho > \rho_D$.

Under bilateral retaliation, the average-protection effects on the belligerents' bilateral rate cancel, leaving that rate unchanged. Against a bystander, however, the relative-treatment effects reinforce each other. The bystander threshold therefore falls from $N\rho_D$ under a unilateral tariff to ρ_D. When $\rho > \rho_D$, both belligerents depreciate against the bystanders and in effective terms.

A multilateral trade war produces the same threshold but shifts the location of adjustment. Under bilateral retaliation, the belligerents' bilateral rate is unchanged and adjustment occurs against the bystanders. In a multilateral trade war there are no untreated bystanders, so all of the tariffing country's bilateral rates move together. In both cases, retaliation increases the importance

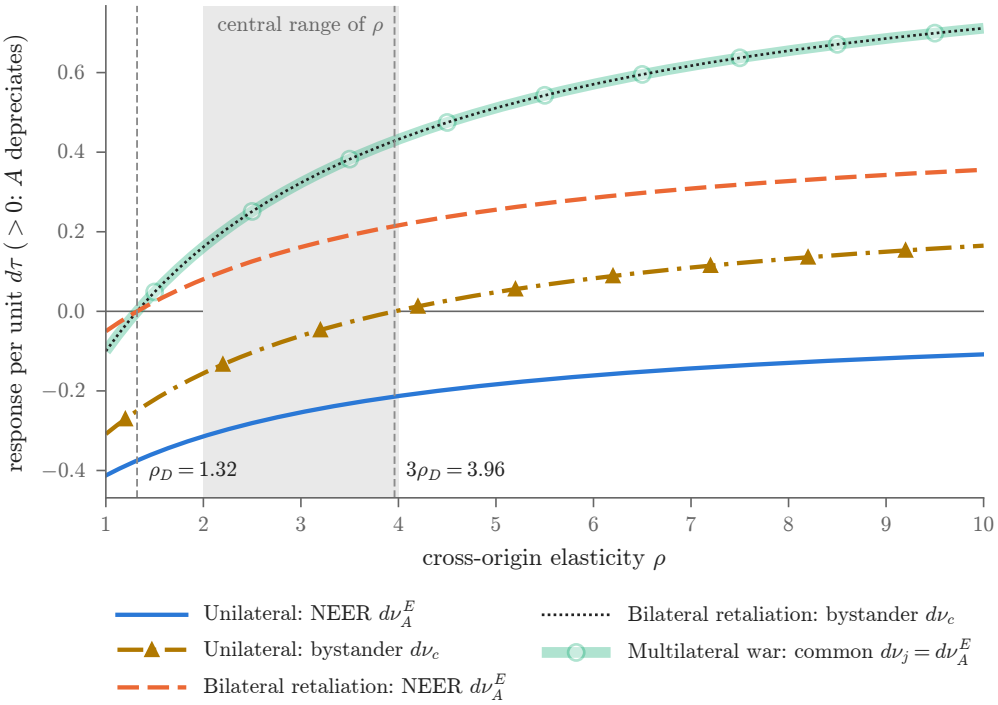


Figure 4: Tariff configurations and exchange-rate responses. Retaliation lowers the bystander-depreciation threshold from $3\rho_D$ to ρ_D when $N = 3$. *Notes:* US-like calibration; shading denotes $\rho \in [2, 4]$.

of relative treatment relative to average protection. At the US-like calibration, $\rho_D = 1.32$, below the illustrative range $\rho \in [2, 4]$ shown in Fig. 4.

4.2 Connection to Preferential Trade

The bystander response is also a third-country terms-of-trade effect of discriminatory protection, connecting the analysis to the preferential-trade literature. In the canonical competing-exporter model, a tariff on one exporter improves the terms of trade of an untariffed exporter because demand is diverted toward it. In the present model, the tariffing country also produces a competing tradable, so diverted expenditure can instead shift toward domestic production.

The bystander therefore gains only when cross-origin substitution is sufficiently strong relative to domestic diversion. The formal comparison with the competing-exporter model, extensions to a positive common external tariff, and results for discrete preferential cuts are reported in the online supplement.

4.3 The 2025 Tariff Episode

The 2025 episode provides a stylized illustration of the tariff configurations considered above. The exercise is neither a calibration nor a causal test. The model is static, has flexible prices, and is evaluated around balanced trade, while observed exchange rates are high-frequency asset prices affected by many contemporaneous shocks. We therefore use the model only to compare the

Table 1: Exchange-rate changes within the three phases of the 2025 episode (percent; positive = dollar depreciation)

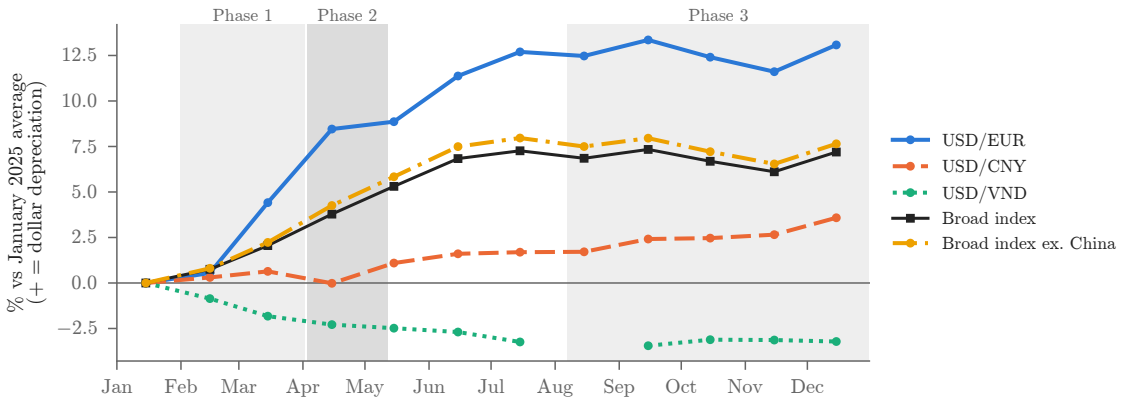
	Phase 1 Feb 1–Apr 2		Phase 2 Apr 3–May 12		Phase 3 Aug 7–Dec 31	
	Monthly	Daily	Monthly	Daily	Monthly	Daily
EUR	4.4	4.5	3.9	2.2	0.3	0.8
CNY	0.6	−0.3	−0.7	0.9	1.9	2.7
CAD	0.2	1.0	2.7	2.2	−0.7	0.2
MXN	1.6	0.6	1.0	4.5	3.4	3.5
VND	−1.8	-2.1^a	−0.5	-0.5^a	0.0	0.5^a
Broad index	2.1	1.7	1.7	2.3	−0.1	0.6
Broad index ex. China	2.2	1.9	2.0	2.5	−0.3	0.4

Notes: Monthly columns compare monthly averages (January with March, March with April, July with December); daily columns compare the last available observations at the phase boundaries. VND daily entries (marked ^{a}) use BIS end-of-month rates. Sources: Federal Reserve H.10 and BIS; see Appendix C.3.

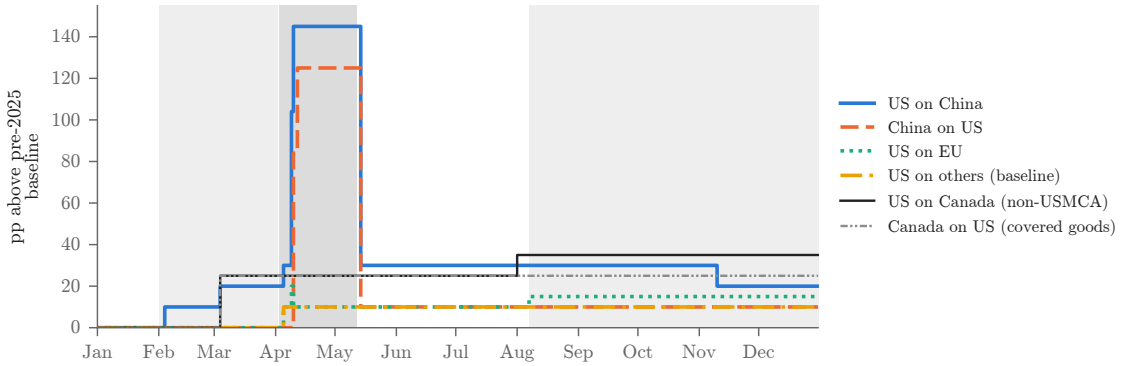
conditional sign patterns associated with different tariff configurations with the observed exchange-rate movements. Observed exchange-rate movements do not identify ρ, ρ_D, or the quantitative contribution of the trade mechanism. Table 1 and Fig. 5 report the data; the Federal Reserve's broad dollar index is the closest counterpart of the model's NEER, and the index excluding China isolates the bystanders.

Phase 1: China-focused escalation (February 1–April 2). The United States raised tariffs on China, while China responded with narrower product-specific tariffs. The United States also imposed tariffs on Canada and Mexico, and Canada retaliated, so this phase is not a clean unilateral-tariff experiment. Approximating it as a tariff concentrated on China, Proposition 3.2 associates the configuration with dollar appreciation against the target and, when $\rho > N\rho_D$, depreciation against untariffed bystanders. The dollar was approximately unchanged against the renminbi and weakened against the euro and the broad index excluding China, matching the bystander sign associated with $\rho > N\rho_D$, although the mixed tariff configuration prevents a sharper comparison.

Phase 2: US–China retaliation (April 3–May 12). The United States and China raised tariffs sharply against each other, while most other partners faced a common 10 percent US baseline tariff. The configuration is therefore closest to bilateral retaliation with an additional uniform component. Proposition 4.1 associates bilateral retaliation with a stable rate between the belligerents and, when $\rho > \rho_D$, depreciation against bystanders and in effective terms. During the phase, the dollar moved little against the renminbi but depreciated against the euro, the broad index excluding China, and the full broad index. This is the phase most closely aligned with the retaliation sign pattern. The comparison establishes sign consistency only; it does not identify the relevant elasticity or the contribution of the trade mechanism.



(a) Exchange rates, percent deviation from the January 2025 average (positive = dollar depreciation)



(b) Headline tariff increases above the pre-2025 baseline (percentage points)

Figure 5: Exchange rates and headline tariff increases during 2025. Shading denotes the three policy phases. Sources: Federal Reserve, BIS, Federal Register, Department of Finance Canada, and China's State Council Tariff Commission.

Phase 3: Near-uniform tariffs with residual retaliation (August 7–December 31). US tariffs became more nearly uniform across partners, although higher rates and retaliation remained for some countries. The broader uniform component strengthens the average-protection channel, which pushes toward effective dollar appreciation, while the remaining discriminatory and retaliatory components can push in the opposite direction. The broad dollar index was approximately flat within the phase, with no clear effective appreciation. Over 2025 as a whole, the dollar depreciated substantially, with most of that movement occurring before August. The static trade mechanism therefore does not account for the full effective depreciation.

The comparison across phases is more informative than any individual episode. A configuration dominated by bilateral retaliation is associated with stronger bystander and effective depreciation forces, while broader uniform protection strengthens the average-protection force. The observed movements in Phase 2 have the sign pattern associated with retaliation, whereas Phase 3 provides no corresponding broad appreciation as protection became more uniform. These comparisons illustrate the importance of the tariff configuration without identifying tariffs as the cause of the exchange-rate movements.

5 Asymmetry

Partner symmetry makes the decomposition between average protection and relative treatment especially transparent. We now ask which results survive when countries differ in size, trade shares, and preferences. The central distinction is between uniform and discriminatory protection. Uniform protection retains the conventional local aggregate result under general asymmetry, while discriminatory protection need not.

5.1 Uniform and Discriminatory Protection

Let b_{Aj} denote partner j's share in A's baseline import expenditure, and let $R_j^B \equiv \partial \nu_j / \partial \tau_{AB}$ be the response of A's bilateral rate against j to a marginal tariff on B. Define

$$R_{A|B} \equiv \sum_{j \neq A} b_{Aj} R_j^B. \tag{5.1}$$

Because producer prices are fixed in their own currencies, a tariff on B improves A's import-weighted aggregate terms of trade if and only if $R_{A|B} < 0$. The welfare result is stated at a balanced free-trade baseline of the model of Section 3.1 with interior shares, lump-sum rebate of tariff revenue, and exchange rates differentiable in tariffs, which holds whenever J is nonsingular.

Lemma 5.1 (First-order welfare)**.** *Under these assumptions, let $d\tau$ be any marginal tariff vector, imposed by A or by other countries. To first order, the change in the welfare of A's representative household, in units of A's income, is*

$$\frac{dU_A}{U_A' \, I_A} = -\, m_A \sum_{j \neq A} b_{Aj} \, d\nu_j, \tag{5.2}$$

where m_A is A's import share of income. A marginal tariff change therefore raises A's own welfare to first order if and only if it improves A's import-weighted aggregate terms of trade.

At free trade, the first-order tariff-revenue gain offsets the direct increase in import prices; the proof is in Appendix B.

Proposition 5.2 (Uniform protection)**.** *At any balanced free-trade baseline satisfying MML and the assumptions of Lemma 5.1, with country sizes, origin shares, home biases and parameters free to differ, a marginal uniform tariff by A on all partners, $d\tau_{Aj} = dt > 0$ for all $j \neq A$, appreciates A's currency weakly against every partner, strictly if A imports from all of them or J is irreducible, improves A's import-weighted aggregate terms of trade, and raises A's welfare to first order.*

Uniform protection leaves relative import prices across foreign suppliers unchanged. Cross-origin substitution therefore creates no differential incidence across partners. Under MML, the tariffing country appreciates against every partner, its aggregate terms of trade improve, and its welfare rises to first order. Asymmetry alone therefore does not overturn the conventional local result. The possibility of reversal arises from discriminatory protection.

Now let A tariff a single target B, and define the import-weighted average response against the remaining suppliers as $\bar{R}^B_{-B} \equiv \sum_{j \neq A,B} \frac{b_{Aj}}{1-b_{AB}} R^B_j$. Then

$$R_{A|B} = b_{AB} R^B_B + (1 - b_{AB}) \bar{R}^B_{-B}. \tag{5.3}$$

If the tariffing country appreciates against the target but depreciates on average against the bystanders, its aggregate terms of trade deteriorate only when the adverse bystander movement is sufficiently large relative to the favorable target movement and the bystanders carry sufficient weight in the import basket. A bilateral bystander reversal is therefore necessary but not sufficient for an aggregate reversal.

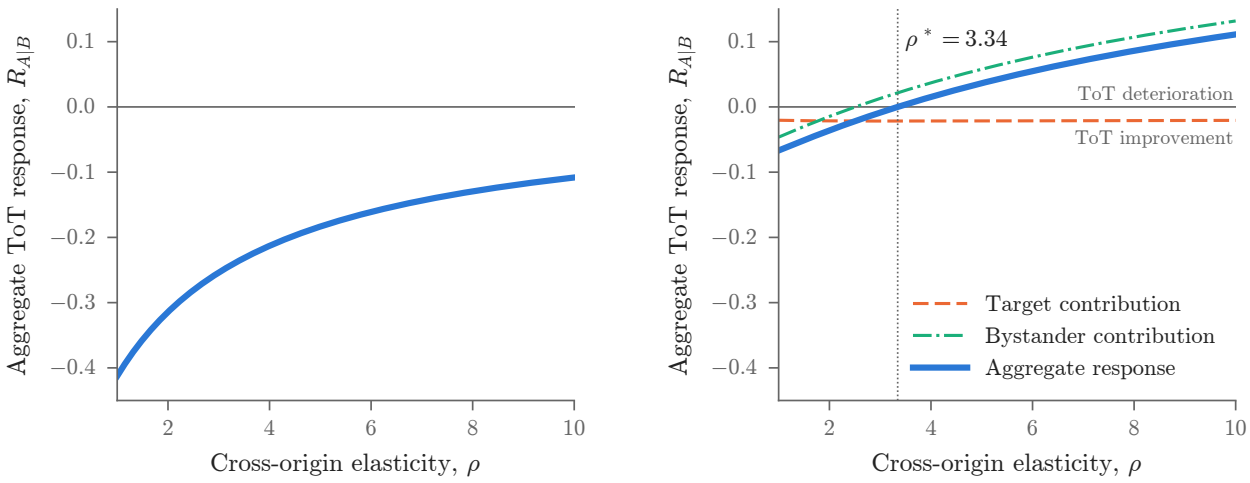
5.2 A Small Target and Large Bystanders

Consider a small or lightly weighted target and relatively large untariffed bystanders. An appreciation against the target then contributes relatively little to the aggregate import price, while depreciation against the bystanders receives greater weight. Strong cross-origin substitution reinforces this asymmetry by redirecting expenditure toward the larger untariffed suppliers.

Proposition 5.3 (Aggregate reversal under discriminatory protection)**.** *Fix a balanced free-trade baseline satisfying MML. Under asymmetry, a marginal discriminatory tariff can worsen the tariffing country's aggregate terms of trade and reduce its welfare to first order.*

Figure 6 shows this weighted-incidence mechanism directly. Under symmetry, the aggregate terms-of-trade response remains favorable. With a small, low-weight target, stronger cross-origin substitution increases the adverse bystander contribution until it dominates the favorable target contribution.

The bilateral, effective, and aggregate terms-of-trade responses need not reverse at the same value of ρ. The NEER uses total-trade weights, while the aggregate terms of trade uses import weights. With heterogeneous bystanders, there is also generally no exact representative-bystander aggregation. The corresponding formulas and additional asymmetric cases are reported in the online supplement.



(a) Symmetric benchmark

(b) Small, low-weight target

Figure 6: Aggregate terms-of-trade response to a discriminatory tariff. Under symmetry, the tariff improves the aggregate terms of trade throughout the range shown. With a small, low-weight target, stronger cross-origin substitution increases the adverse bystander contribution until the aggregate response reverses at $\rho^* = 3.34$.

6 Conclusion

With two countries, a tariff moves a single exchange rate, so the bilateral exchange rate, the effective exchange rate, and the long-run terms of trade coincide. In a multilateral economy tariffs change both average protection and relative treatment across foreign suppliers, so bilateral rates, the effective exchange rate, and the aggregate terms of trade need not move together, and the resulting exchange-rate adjustment depends on the full vector of trade-balance disturbances rather than on the tariffing country's bilateral relationship with the target alone.

This distinction changes the conventional tariff result. Under symmetry, a unilateral tariff appreciates the tariffing country's effective exchange rate but can depreciate its currency against untariffed bystanders when cross-origin substitution is sufficiently strong relative to domestic diversion. Retaliation strengthens the relative-treatment force and lowers the bystander threshold. Under asymmetry, uniform protection retains the conventional local aggregate terms-of-trade and welfare result, while discriminatory protection can overturn it. In our numerical comparative statics, such reversals arise most readily when the target is small or lightly weighted and cross-origin substitution toward larger bystanders is sufficiently strong.

The 2025 episode illustrates why the cross-country configuration of protection matters, but the sign comparisons do not identify the trade mechanism or its quantitative contribution. The broader implication is that the effect of tariffs on "the exchange rate" is not well defined without specifying both the exchange-rate object and the configuration of tariffs across trading partners.

References

Amiti, Mary, Stephen J. Redding, and David E. Weinstein, "The Impact of the 2018 Tariffs on Prices and Welfare," *Journal of Economic Perspectives*, 2019, *33* (4), 187–210.

Arrow, Kenneth J. and Leonid Hurwicz, "On the Stability of the Competitive Equilibrium, I," *Econometrica*, 1958, *26* (4), 522–552.

_ , H. D. Block, and Leonid Hurwicz, "On the Stability of the Competitive Equilibrium, II," *Econometrica*, 1959, *27* (1), 82–109.

Bagwell, Kyle and Robert W. Staiger, "An Economic Theory of GATT," *American Economic Review*, 1999, *89* (1), 215–248.

_ and _ , *The Economics of the World Trading System*, Cambridge, MA: MIT Press, 2002.

Barattieri, Alessandro, Matteo Cacciatore, and Fabio Ghironi, "Protectionism and the Business Cycle," *Journal of International Economics*, 2021, *129*, 103417.

Bergin, Paul R. and Giancarlo Corsetti, "The Macroeconomic Stabilization of Tariff Shocks: What Is the Optimal Monetary Response?," *Journal of International Economics*, 2023, *143*, 103758.

Bond, Eric W. and Constantinos Syropoulos, "The Size of Trading Blocs: Market Power and World Welfare Effects," *Journal of International Economics*, 1996, *40* (3–4), 411–437.

Broda, Christian and David E. Weinstein, "Globalization and the Gains from Variety," *Quarterly Journal of Economics*, 2006, *121* (2), 541–585.

Cavallo, Alberto, Gita Gopinath, Brent Neiman, and Jenny Tang, "Tariff Pass-Through at the Border and at the Store: Evidence from US Trade Policy," *American Economic Review: Insights*, 2021, *3* (1), 19–34.

Dornbusch, Rudiger, "Tariffs and Nontraded Goods," *Journal of International Economics*, 1974, *4* (2), 177–185.

Eichengreen, Barry J., "A Dynamic Model of Tariffs, Output and Employment under Flexible Exchange Rates," *Journal of International Economics*, 1981, *11* (3), 341–359.

Erceg, Christopher, Andrea Prestipino, and Andrea Raffo, "Trade Policies and Fiscal Devaluations," *American Economic Journal: Macroeconomics*, 2023, *15* (4), 104–140.

Fajgelbaum, Pablo D., Pinelopi K. Goldberg, Patrick J. Kennedy, and Amit K. Khandelwal, "The Return to Protectionism," *Quarterly Journal of Economics*, 2020, *135* (1), 1–55.

Feenstra, Robert C., Philip Luck, Maurice Obstfeld, and Katheryn N. Russ, "In Search of the Armington Elasticity," *Review of Economics and Statistics*, 2018, *100* (1), 135–150.

Furceri, Davide, Swarnali A. Hannan, Jonathan D. Ostry, and Andrew K. Rose, "Macroeconomic Consequences of Tariffs," IMF Working Paper 19/9, International Monetary Fund 2019.

Hicks, John R., *Value and Capital: An Inquiry into Some Fundamental Principles of Economic Theory*, Oxford: Clarendon Press, 1939.

Itskhoki, Oleg and Dmitry Mukhin, "The Optimal Macro Tariff," NBER Working Paper 33839, National Bureau of Economic Research 2025.

Jeanne, Olivier and Jeongwon Son, "To What Extent Are Tariffs Offset by Exchange Rates?," *Journal of International Money and Finance*, 2024, *142*, 103015.

Kalemli-Özcan, Şebnem, Can Soylu, and Muhammed A. Yıldırım, "Global Trade, Tariff Uncertainty, and the US Dollar," *AEA Papers and Proceedings*, 2026, *116*, 47–52. Longer version: "Tariff Uncertainty and the U.S. Dollar," NBER Working Paper 34728, 2026; also CEPR Discussion Paper 21062.

Krugman, Paul, "The Macroeconomics of Protection with a Floating Exchange Rate," *Carnegie-Rochester Conference Series on Public Policy*, 1982, *16*, 141–182.

Lerner, Abba P., "The Symmetry between Import and Export Taxes," *Economica*, 1936, *3* (11), 306–313.

Li, Wei, Bin Qiu, Zi Wang, Yanbo Zeng, and Yuan Zi, "Armington Elasticities and the Third-Country Effects of Trade Wars," CEPR Discussion Paper 17409, Centre for Economic Policy Research 2022.

Lindé, Jesper and Andrea Pescatori, "The Macroeconomic Effects of Trade Tariffs: Revisiting the Lerner Symmetry Result," *Journal of International Money and Finance*, 2019, *95*, 52–69.

Matveev, Dmitry and Francisco J. Ruge-Murcia, "Tariffs and the Exchange Rate: Evidence from Twitter," Staff Working Paper 2021-36, Bank of Canada 2021. Published in *IMF Economic Review* 72(3), 2024, 1185–1211.

Metzler, Lloyd A., "Stability of Multiple Markets: The Hicks Conditions," *Econometrica*, 1945, *13* (4), 277–292.

_ , "Tariffs, the Terms of Trade, and the Distribution of National Income," *Journal of Political Economy*, 1949, *57* (1), 1–29.

Mosak, Jacob L., *General-Equilibrium Theory in International Trade* number 7. In 'Cowles Commission Monograph.', Bloomington, IN: Principia Press, 1944.

Mundell, Robert A., "Flexible Exchange Rates and Employment Policy," *Canadian Journal of Economics and Political Science*, 1961, *27* (4), 509–517.

_ , "Tariff Preferences and the Terms of Trade," *Manchester School of Economic and Social Studies*, 1964, *32* (1), 1–13.

Ostry, Daniel, Simon Lloyd, and Giancarlo Corsetti, "Trading Blows: The Exchange-Rate Response to Tariffs and Retaliations," Staff Working Paper 1139, Bank of England 2025. Also CEPR Discussion Paper 20452.

Viner, Jacob, *The Customs Union Issue*, New York: Carnegie Endowment for International Peace, 1950.

Werning, Iván, Guido Lorenzoni, and Veronica Guerrieri, "Tariffs as Cost-Push Shocks: Implications for Optimal Monetary Policy," NBER Working Paper 33772, National Bureau of Economic Research 2025.

A The Linearized System

This appendix derives the linear system Eq. (2.2) of the nested-CES model of Section 3.1 and evaluates it at the symmetric benchmark.

A.1 General linearization

All derivatives are first-order and evaluated at a balanced free-trade baseline, $\tau_{ij} = 0$ and $\mathrm{TB}_i = 0$ for every $i \in \mathcal{I}$. Values are in the common unit of account of Section 2.1, with $v_A = 1$. Write $\chi_l = \log v_l$ for $l \in \mathcal{I}$; then $\nu_j = \log e_{Aj} = \chi_j - \chi_A$, $\nu_A \equiv 0$, and $d\log e_{ij} = d\nu_j - d\nu_i$.

The baseline is described by origin shares b_{ij} within i's import bundle ($b_{ii} = 0$, $\sum_{j\neq i} b_{ij} = 1$), home shares h_i of i's tradable spending, and incomes $y_i \equiv \tilde{I}_i$ in the common unit. The share of i's income spent on j's good is $s_{ij} = \alpha_T^i(1-h_i)b_{ij}$, and the border value of i's imports from j is $f_{ij} = s_{ij}y_i$; η_i, ρ_i and α_T^i may differ by country. Log-differentiating the price indices of Section 3.1, the demand system Eq. (3.1), income Eq. (3.2) and the trade balance Eq. (3.3) gives

$$\begin{aligned}
dp_{ij} &= d\nu_j - d\nu_i + d\tau_{ij}, \qquad dp_{ii} = 0,\\
d\log P_{Mi} &= \textstyle\sum_{j\neq i} b_{ij}\, dp_{ij}, \qquad d\log P_{Ti} = (1-h_i)\, d\log P_{Mi},\\
d\log s_{ii} &= -(1-\eta_i)(1-h_i)\, d\log P_{Mi},\\
d\log s_{ij} &= (1-\rho_i)(dp_{ij} - d\log P_{Mi}) + (1-\eta_i)h_i\, d\log P_{Mi},\\
d\log \tilde{I}_i &= d\nu_i + \textstyle\sum_{j\neq i} s_{ij}\, d\tau_{ij},\\
d\mathrm{TB}_i &= \textstyle\sum_{k\neq i} f_{ki}(d\log s_{ki} + d\log \tilde{I}_k - d\tau_{ki}) - \sum_{j\neq i} f_{ij}(d\log s_{ij} + d\log \tilde{I}_i - d\tau_{ij}),
\end{aligned} \tag{A.1}$$

where dp_{ij} is the log change in i's consumer price of j's good in i's currency. At free trade, only the direct revenue term survives in $d\log \tilde{I}_i$, and $d\log f_{ij} = d\log s_{ij} + d\log \tilde{I}_i - d\tau_{ij}$. Stacking the trade-balance equations gives the matrices J and G in Eq. (2.2).

A.2 The symmetric benchmark

At the symmetric benchmark the country subscripts on η, ρ and α_T are dropped; $b_{ij} = 1/(N-1)$, $h_i = \alpha_D$, $y_i = 1$, and $f_{ij} = s_{ij} = m/(N-1)$ with $m = \alpha_T(1-\alpha_D)$.

Lemma A.1 (Jacobian at the symmetric baseline)**.** *At the symmetric balanced baseline, $\widetilde{J}_{il} = mM_N/(N-1)$ for $i,l \in \mathcal{I}$ with $l \neq i$, and $\widetilde{J}_{ii} = -mM_N$. The reduced matrix $J = (\widetilde{J}_{il})_{i,l\in\mathcal{I}_{-A}}$ satisfies*

$$-J = j_S\,(NI - \mathbf{1}\mathbf{1}'), \qquad j_S = \frac{mM_N}{N-1}, \qquad j_D = Nj_S. \tag{A.2}$$

Proof. Substituting the symmetric baseline into Eq. (A.1) gives $\widetilde{J}_{il} = mM_N/(N-1)$ for $i \neq l$. Homogeneity and balanced trade imply zero row sums, giving $\widetilde{J}_{ii} = -mM_N$. Deleting the numeraire row and column yields Eq. (A.2). The corresponding tariff disturbance follows from Eq. (A.1) at $d\nu = 0$, giving Eq. (A.3) and the eigenvalues in Eq. (A.4). Detailed differentiation is reported in the online supplement. □

At $d\nu = 0$, a change $t = (t_j)_{j\in\mathcal{I}_{-A}}$, $t_j = d\tau_{Aj}$, in A's tariffs moves only A's import prices and income, so among A's partners only i's exports to A change:

$$d\mathrm{TB}_i\big|_{\nu} = f_{Ai}(d\log s_{Ai} + d\log \tilde{I}_A - t_i) = \frac{m}{N-1}\Big[-\rho_D\,\bar{t} - \rho\,(t_i - \bar{t})\Big], \tag{A.3}$$

using $d\log P_{MA} = \bar{t}$, $d\log s_{Ai} = (1-\rho)(t_i-\bar{t})+(1-\eta)\alpha_D\bar{t}$, $d\log \tilde{I}_A = m\bar{t}$ and $1-(1-\eta)\alpha_D-m = \rho_D$. Equivalently, $G^{(A)} = \frac{m}{N-1}\big[-\rho I + \frac{\rho-\rho_D}{N-1}\mathbf{1}\mathbf{1}'\big]$, with eigenvalues

$$g_S = -\frac{m\rho_D}{N-1}, \qquad g_D = -\frac{m\rho}{N-1}, \tag{A.4}$$

on span$\{\mathbf{1}\}$ and on the zero-sum subspace.

A.3 The two-country case

At $N = 2$ the margin $(1-\rho)$ is inactive, $dp_{AB} = d\log P_{MA} = d\nu_B + d\tau$ and $dp_{BA} = d\log P_{MB} = -d\nu_B$, so

$$d\log s_{AB} = (1-\eta)\alpha_D(d\nu_B + d\tau), \qquad d\log s_{BA} = -(1-\eta)\alpha_D\,d\nu_B,$$
$$d\log \tilde{I}_A = m\,d\tau, \qquad d\log \tilde{I}_B = d\nu_B.$$

Substituting into $\mathrm{TB}_A = f_{BA} - f_{AB}$,

$$d\mathrm{TB}_A = m\Big[\big(1+2\alpha_D(\eta-1)\big)d\nu_B + \big(1-m+\alpha_D(\eta-1)\big)d\tau\Big] = mM_2\,d\nu_B + m\rho_D\,d\tau,$$

with $M_2 = 1+2\alpha_D(\eta-1)$ and $\rho_D = 1-m+\alpha_D(\eta-1) = \alpha_D\eta + (1-\alpha_D)(1-\alpha_T)$. Setting $d\mathrm{TB}_A = 0$ gives Eq. (3.4).

At the balanced free-trade baseline, the corresponding import and export elasticities satisfy $\varepsilon_X + \varepsilon_M - 1 = M_2$, so $M_2 > 0$ is the Marshall–Lerner condition used in Section 3.2.

A.4 Unilateral tariffs

Here $N \geq 3$, the baseline is the symmetric benchmark, and $M_N > 0$. For a unilateral tariff, $t = d\tau\, e_B$, and an individual bystander c, $\bar{t} = d\tau/(N-1)$, $\widetilde{t}_B = \frac{N-2}{N-1}d\tau$ and $\widetilde{t}_c = -\frac{d\tau}{N-1}$. By Eq. (3.7),

$$d\nu_B = \frac{\lambda_{\mathrm{avg}} + (N-2)\lambda_{\mathrm{rel}}}{N-1}\,d\tau = -\frac{N\rho_D + (N-2)\rho}{N(N-1)M_N}\,d\tau,$$
$$d\nu_c = \frac{\lambda_{\mathrm{avg}} - \lambda_{\mathrm{rel}}}{N-1}\,d\tau = \frac{\rho - N\rho_D}{N(N-1)M_N}\,d\tau, \qquad d\nu_A^E = \frac{\lambda_{\mathrm{avg}}}{N-1}\,d\tau = -\frac{\rho_D}{(N-1)M_N}\,d\tau,$$

which is Eq. (3.8); $d\nu_c > 0$ if and only if $\rho > N\rho_D$. By Eq. (A.3), the target's and each bystander's disturbances per unit $d\tau$ are

$$F_B = -\frac{m[\rho_D + (N-2)\rho]}{(N-1)^2} < 0, \qquad F_c = \frac{m(\rho-\rho_D)}{(N-1)^2},$$

the second of which is Eq. (3.10). For a general tariff vector with $\bar{t} > 0$, substituting λ_{avg} and λ_{rel} into $d\nu_j = \lambda_{\text{avg}}\bar{t} + \lambda_{\text{rel}}(t_j - \bar{t})$ shows that A depreciates against partner j if and only if

$$\frac{\rho}{N}(\bar{t} - t_j) > \rho_D\,\bar{t} \qquad \Longleftrightarrow \qquad \frac{t_j}{\bar{t}} < 1 - \frac{N\rho_D}{\rho}. \tag{A.5}$$

If A raises its tariff by $d\tau$ on $k \le N-2$ partners, its response against each untargeted partner is k times the unilateral bystander response; the derivation is in the online supplement.

B Proofs

Notation and the sign convention are those of the main text; $m \equiv \alpha_T(1-\alpha_D) > 0$ at the symmetric benchmark. Unless stated otherwise, derivatives are first-order and evaluated at a balanced free-trade baseline, using the linear system of Appendix A.

B.1 Multilateral adjustment and symmetric partners

Proof of Lemma 2.2. Free trade is not required. By homogeneity and Euler's theorem, $\sum_{l\in\mathcal{I}} \widetilde{J}_{il} = \text{TB}_i = 0$ at the balanced baseline. Since $\nu_l = \chi_l - \chi_A$ with χ_A fixed, J is $\widetilde{J}$ with row and column A deleted. Its off-diagonal entries are nonnegative, and for each $i \in \mathcal{I}_{-A}$

$$|J_{ii}| - \sum_{l\in\mathcal{I}_{-A},\, l\neq i} J_{il} = \widetilde{J}_{iA} \ge 0,$$

strictly in at least one row. The diagonal of $-J$ is strictly positive: for $N = 2$, $|J_{BB}| = \widetilde{J}_{BA} > 0$; for $N \ge 3$, $J_{ii} = 0$ would force the nonnegative off-diagonal entries of row i to vanish, contradicting irreducibility. Write $-J = D - K$ with $D = \text{diag}(-J) > 0$ and $K \ge 0$. The matrix $D^{-1}K$ is nonnegative and irreducible, with every row sum at most one and at least one strictly below one. The spectral radius of an irreducible nonnegative matrix is at most its largest row sum, with equality only if all row sums are equal (Perron–Frobenius), so the spectral radius of $D^{-1}K$ is below one and

$$(-J)^{-1} = (I - D^{-1}K)^{-1}D^{-1} = \sum_{k\ge 0}(D^{-1}K)^k D^{-1} \ge 0.$$

Hence MML holds. Because $D^{-1}K$ is irreducible, the series is entrywise positive, and $\det(-J) = \det D \cdot \det(I - D^{-1}K) > 0$. □

Proof of Proposition 2.3. No demand system, free-trade baseline or stability condition is used. If $PMP' = M$ for every $(N-1)\times(N-1)$ permutation matrix P, conjugating by permutations shows that all diagonal entries are equal and all off-diagonal entries are equal, so $M = aI + b\mathbf{1}\mathbf{1}'$, with eigenvalue $a + (N-1)b$ on $\text{span}\{\mathbf{1}\}$ and a on the zero-sum subspace. For $-J = a_J I + b_J \mathbf{1}\mathbf{1}'$ and $G^{(A)} = a_G I + b_G \mathbf{1}\mathbf{1}'$ this gives $j_S = a_J + (N-1)b_J$, $j_D = a_J$, $g_S = a_G + (N-1)b_G$ and $g_D = a_G$,

and for $j_S, j_D \neq 0$

$$(-J)^{-1} = \frac{1}{j_D} I + \frac{1}{N-1}\Big(\frac{1}{j_S} - \frac{1}{j_D}\Big)\mathbf{1}\mathbf{1}'.$$

Thus $(-J)^{-1}$ and $G^{(A)}$ share the two eigenspaces, and applying them to $t = \bar{t}\mathbf{1} + \widetilde{t}$ gives Eq. (2.6). Partner symmetry makes the baseline flows $f_{Aj} + f_{jA}$ equal across j, so $w_{Aj} = 1/(N-1)$; since $\widetilde{t}$ sums to zero, $d\nu_A^E = \lambda_{\text{avg}}\bar{t}$, and the other two parts of Eq. (2.7) follow directly.

For the MML characterization, the off-diagonal entry of J is $-b_J = (j_D - j_S)/(N-1)$, so the sign pattern holds if and only if $j_D \geq j_S$. If $j_D \geq j_S > 0$, $-J$ is nonsingular and every entry of $(-J)^{-1}$ above is nonnegative, so MML holds. Conversely, under MML the sign pattern gives $j_D \geq j_S$, nonsingularity gives $j_S \neq 0$, and $(-J)^{-1}\mathbf{1} = \mathbf{1}/j_S \geq 0$ gives $j_S > 0$. □

B.2 The symmetric nested-CES model

The baseline is the symmetric benchmark of Section 3.1 with $N \geq 3$; for $N = 2$, Theorem 3.1 reduces to Eq. (3.4).

Proof of Theorem 3.1. The benchmark is invariant to relabeling A's partners, so Proposition 2.3 applies, and the eigenvalues Eq. (3.6) are those of Lemma A.1 and Eq. (A.4). Since $m > 0$, $j_S = mM_N/(N-1) > 0$ if and only if $M_N > 0$, and then $j_D = Nj_S > j_S$. By Proposition 2.3, MML is therefore equivalent to $M_N > 0$; if $M_N \leq 0$, then $j_S \leq 0$ and MML fails. If $\eta \geq 1$, the bracket in Eq. (3.5) is 1 plus a nonnegative term plus $(N-2)\rho > 0$, so $M_N > 0$. Given $M_N > 0$, dividing Eq. (A.4) by the eigenvalues of $-J$ gives $\lambda_{\text{avg}} = -\rho_D/M_N$ and $\lambda_{\text{rel}} = -\rho/(NM_N)$, and Eq. (3.7) is Eq. (2.6). □

Proof of Proposition 3.2. Equation (3.8) follows from Eq. (3.7) (Appendix A.4). Under MML $M_N > 0$, and $\rho_D > 0$ since $\alpha_D\eta > 0$; hence $d\nu_A^E/d\tau < 0$ and, since $N\rho_D + (N-2)\rho > 0$, $d\nu_B/d\tau < 0$. The denominator of $d\nu_c/d\tau$ is positive, so $d\nu_c/d\tau > 0$ if and only if $\rho > N\rho_D$, with $d\nu_c = 0$ at $\rho = N\rho_D$. For (iv), with $k \leq N-2$ targets the response against each untargeted partner is k times the unilateral response (Appendix A.4), so the threshold is again $\rho > N\rho_D$. For a general tariff vector with $\bar{t} > 0$, Eq. (A.5) gives the stated condition; when $t_j \geq 0$ its left side is nonnegative, so it can hold only if $1 - N\rho_D/\rho > 0$, that is, $\rho > N\rho_D$, and it then implies $t_j < \bar{t}$. □

Proof of Corollary 3.3. Since $N\rho_D = N[1 + \alpha_D(\eta - 1) - \alpha_T(1 - \alpha_D)]$, Eq. (3.9) is an identity; under $\alpha_D \geq 1/N$ both terms on its right are nonnegative, so $N\rho_D \geq \eta$, and since a bystander depreciation requires $\rho > N\rho_D$ (Proposition 3.2), it requires $\rho > \eta$. With $\rho = \eta = \sigma$, Eq. (3.8) gives $d\nu_c/d\tau = (\sigma - N\rho_D)/[N(N-1)M_N]$ with $\sigma - N\rho_D = -[\sigma(N\alpha_D - 1) + N(1-\alpha_D)(1-\alpha_T)] \leq 0$; MML is used only to give $M_N > 0$, so $d\nu_c/d\tau \leq 0$. Since $\sigma > 0$ and $\alpha_D < 1$, the bracket vanishes if and only if $\alpha_D = 1/N$ and $\alpha_T = 1$. At $\alpha_D = 1/N$, $N\rho_D - \sigma = (N-1)(1-\alpha_T)$ and $M_N = \sigma > 0$, so MML holds for every $\sigma > 0$, and $d\nu_c/d\tau = -(1-\alpha_T)/(N\sigma)$, which is negative and tends to zero as $\sigma \to \infty$ if $\alpha_T < 1$ and is identically zero if $\alpha_T = 1$. □

Proposition 3.4 applies to any marginal tariff change that is invariant to permutations of the $N-2$ bystanders and leaves tariffs among bystanders unchanged; the composite's tariffs are the common bystander values. In the three-region economy, C has income $y_C = N-2$, home share h_C with $(N-1)h_C = N-3+2\alpha_D$, weight $\frac{1}{2}$ on each of A and B in its import nest, tradable share α_T, cross-origin elasticity ρ and home–import elasticity η_C; A and B place import weight $\frac{1}{N-1}$ on each other and $\frac{N-2}{N-1}$ on C.

Proof of Proposition 3.4. The argument compares the two linearized systems at the benchmark; it does not claim that the nonlinear economies coincide at finite tariffs.

Common bystander response. By Lemma A.1, $-J = j_S(NI - \mathbf{1}\mathbf{1}')$ with $j_S = mM_N/(N-1) \neq 0$, so Eq. (2.2) has a unique solution. Baseline and shock are invariant to permutations of the bystanders, hence so is the solution: $d\nu_c = d\nu_C$ for all c. The equilibrium is then given by $d\mathrm{TB}_B = 0$ and $\sum_c d\mathrm{TB}_c = 0$ in $(d\nu_B, d\nu_C)$.

Composite structure. Summing Eq. (3.3) over bystanders, intra-bloc flows cancel, so $\sum_c \mathrm{TB}_c$ has the form of $\mathrm{TB}_C = f_{AC} + f_{BC} - f_{CA} - f_{CB}$. Since all bystander goods face a common price change in A and in B, A's and B's demand for the bloc's goods responds as their demand for C's good. It remains to match the bloc's demand for A's and B's goods.

Internal elasticity. Measure prices in bystander currency, so other bystanders' goods have $dp = 0$, and write $S \equiv dp_{cA} + dp_{cB}$ and $D \equiv dp_{cA} - dp_{cB}$. A bystander's import index moves by $S/(N-1)$ and the composite's by $S/2$, so

$$d\log s_{cA} = (1-\rho)\frac{D}{2} + \Big[(1-\rho)\frac{N-3}{2(N-1)} + (1-\eta)\frac{\alpha_D}{N-1}\Big]S,$$
$$d\log s_{CA} = (1-\rho)\frac{D}{2} + (1-\eta_C)h_C\frac{S}{2}.$$

The D terms coincide; the S terms coincide if and only if $(1-\eta_C)(N-1)h_C = (1-\rho)(N-3) + 2\alpha_D(1-\eta)$, which, with $(N-1)h_C = N-3+2\alpha_D$, is Eq. (3.11). The argument for B's good replaces D by $-D$.

Equivalence. The mapped baseline flows and demand responses therefore reproduce the two independent equilibrium conditions for $(d\nu_B, d\nu_C)$. Nonsingularity gives the same unique solution in the original and composite economies, and the composite trade weights reproduce A's NEER. For $N = 3$, C is the single bystander. Additional flow-by-flow verification is reported in the online supplement. □

B.3 Retaliation, welfare, and asymmetry

Proof of Proposition 4.1. Under MML $M_N > 0$ (Theorem 3.1), and the response is linear in tariffs, so responses to simultaneous tariffs superpose. (i) Shock 1, $d\tau_{AB} = d\tau$, is Eq. (3.8). Shock 2, $d\tau_{BA} = d\tau$, is the same experiment with A and B exchanged, so $d\log e_{BA} = -\frac{N\rho_D + (N-2)\rho}{N(N-1)M_N}d\tau$ and

$d\log e_{Bc} = \frac{\rho - N\rho_D}{N(N-1)M_N} d\tau$; with $e_{AB} = 1/e_{BA}$ and $e_{Ac} = e_{AB}e_{Bc}$,

$$d\nu_B^{(2)} = \frac{N\rho_D + (N-2)\rho}{N(N-1)M_N}\, d\tau, \qquad d\nu_c^{(2)} = \frac{\rho}{NM_N}\, d\tau.$$

Adding shock 1, $d\nu_B = 0$ and

$$\frac{d\nu_c}{d\tau} = \frac{\rho - N\rho_D}{N(N-1)M_N} + \frac{\rho}{NM_N} = \frac{\rho - \rho_D}{(N-1)M_N}$$

for every bystander, so cross rates among bystanders are unchanged and $d\log e_{Bc} = d\nu_c - d\nu_B = d\nu_c$. With equal weights, $d\nu_A^E = \frac{N-2}{N-1} d\nu_c$, and B's NEER, $[0 + (N-2)d\nu_c]/(N-1)$, is the same. Since $M_N > 0$ and $N \geq 3$, every response has the sign of $\rho - \rho_D$.

(ii) A's uniform tariff contributes $\lambda_{\text{avg}} d\tau = -\frac{\rho_D}{M_N} d\tau$ to every $d\nu_j$. Partner j's tariff on A is shock 2 with j in the role of B: it contributes $\frac{N\rho_D + (N-2)\rho}{N(N-1)M_N} d\tau$ to $d\nu_j$ and $\frac{\rho}{NM_N} d\tau$ to each of the other $N-2$ partners. Summing,

$$\frac{d\nu_j}{d\tau} = -\frac{\rho_D}{M_N} + \frac{N\rho_D + (N-2)\rho}{N(N-1)M_N} + \frac{(N-2)\rho}{NM_N} = \frac{(N-2)(\rho - \rho_D)}{(N-1)M_N},$$

common to all j, so cross rates among partners are unchanged and $d\nu_A^E$ equals this response, with the sign of $\rho - \rho_D$. □

Proof of Lemma 5.1. Indirect utility is $V_A = \kappa\, I_A / (P_{TA}^{\alpha_T} P_{NA}^{1-\alpha_T})$, so the welfare change in units of income is $d\log I_A - \alpha_T\, d\log P_{TA} - (1-\alpha_T)\, d\log P_{NA}$. By Shephard's lemma, $\alpha_T\, d\log P_{TA} = \sum_j s_{Aj}\, d\log p_{Aj}$. Under the producer-price normalization P_{NA} and the domestic tradable's price are fixed in A's currency, and at free trade $d\log p_{Aj} = d\nu_j + d\tau_{Aj}$. Income is $I_A = w_A L_A / (1 - \sum_j \theta_{Aj} s_{Aj})$ with $\theta_{Aj} = \tau_{Aj}/(1+\tau_{Aj})$; at free trade $\theta_{Aj} = 0$ and $d\theta_{Aj} = d\tau_{Aj}$, so share changes drop out and $d\log I_A = \sum_j s_{Aj}\, d\tau_{Aj}$. Combining,

$$\frac{dU_A}{U_A' I_A} = \sum_j s_{Aj}\, d\tau_{Aj} - \sum_j s_{Aj}(d\nu_j + d\tau_{Aj}) = -m_A \sum_{j \neq A} b_{Aj}\, d\nu_j,$$

with $m_A = \sum_j s_{Aj}$ and $b_{Aj} = s_{Aj}/m_A$. The tariff terms cancel because $\tau_{Aj} = 0$ at the baseline; the cancellation is first order only. Other countries' tariffs enter A's prices and income only through $d\nu$, so the formula holds for any tariff vector. □

Proof of Proposition 5.2. Use Eq. (A.1) with country-specific η_i, ρ_i and α_T^i, so $s_{Aj} = \alpha_T^A(1-h_A)b_{Aj}$ and $m_A = \alpha_T^A(1-h_A)$. At $d\nu = 0$, no country other than A faces a price change, so $F_j = f_{Aj}(d\log s_{Aj} + d\log \tilde{I}_A - dt)$. In A, $dp_{Aj} = dt$ for every j, so $d\log P_{MA} = dt$, the cross-origin term vanishes, $d\log s_{Aj} = (1-\eta_A)h_A\, dt$ and $d\log \tilde{I}_A = m_A\, dt$. Hence

$$F_j = f_{Aj}\big[(1-\eta_A)h_A + \alpha_T^A(1-h_A) - 1\big]dt = -f_{Aj}\,\rho_D^A\, dt, \qquad \rho_D^A \equiv \eta_A h_A + (1-h_A)(1-\alpha_T^A) > 0,$$

so $F_j \leq 0$, strictly if $f_{Aj} > 0$. Under MML, Eq. (2.4) gives $d\nu_i \leq 0$ for every i. If A imports from every partner, $F < 0$, and each row of the nonsingular nonnegative $(-J)^{-1}$ has a positive entry, so $d\nu_i < 0$ for every i; if J is irreducible, $(-J)^{-1} > 0$ entrywise (Lemma 2.2), so $d\nu_i < 0$ for every i whenever $F \neq 0$. By Lemma 5.1, the welfare change $-m_A \sum_j b_{Aj} d\nu_j$ is positive, and the aggregate terms of trade improve for the same reason. □

Proof of Proposition 5.3. By Lemma 5.1, a marginal tariff on B worsens A's aggregate terms of trade and lowers its welfare to first order if and only if $R_{A|B} > 0$. Consider the three-country baseline used in panel (b) of Fig. 6, with the US-like parameters, $L = (1, 0.22, 1)$, $\beta_{AB} = 0.2$, $\beta_{AC} = 0.8$, and the remaining origin weights symmetric. The free-trade baseline is solved at $\bar{\rho} = 3\rho_D = 3.96$ and then held fixed, giving $b_{AB} = 0.094$. MML holds over the range used in the figure. The aggregate response crosses zero at $\rho^* = 3.344$ and is positive above that threshold over the reported range. Hence a discriminatory tariff can worsen A's aggregate terms of trade and reduce its welfare to first order. The general analytical characterization and additional asymmetric configurations are reported in the online supplement. □

B.4 Verification

The analytical Jacobians and tariff disturbances were checked against centered finite differences of the nonlinear model at symmetric and asymmetric baselines. Closed-form responses were also checked against direct solutions of the linear system, with MML verified at every reported threshold.

C Further Results and Data

This appendix states sufficient conditions for the two-country sign result of Section 2.1, derives the real-exchange-rate results discussed in Section 3.5, and documents the 2025 data. Throughout, e_{Aj} is units of A's currency per unit of j's currency, so a rise is a depreciation of A against j.

C.1 Two-country sufficient conditions

Let $e \equiv e_{AB}$ and let τ be A's tariff on imports from B. If $\mathrm{TB}_A(e, \tau)$ is continuously differentiable and $\mathrm{TB}_{A,e} \neq 0$, the implicit-function theorem gives

$$\frac{de}{d\tau} = -\frac{\mathrm{TB}_{A,\tau}}{\mathrm{TB}_{A,e}}.$$

A tariff therefore appreciates A's currency locally when $\mathrm{TB}_{A,\tau} > 0$ and $\mathrm{TB}_{A,e} > 0$. At the balanced baseline, the second condition is the Marshall–Lerner condition $\varepsilon_X + \varepsilon_M > 1$. In the nested-CES model at free trade, $\mathrm{TB}_{A,\tau} > 0$ because the domestic diversion margin ρ_D is strictly positive. These are sufficient conditions rather than universal properties: alternative fiscal closures, pricing

conventions, incomplete pass-through, or endogenous markups can change the underlying quantity responses and either sign.

C.2 Real exchange rates

Let $P_i = P_{Ti}^{\alpha_T} P_{Ni}^{1-\alpha_T}$ be i's consumer price index in its own currency and $q_{Aj} \equiv e_{Aj} P_j / P_A$ the real bilateral exchange rate, so that a rise is a real depreciation of A. At the symmetric benchmark the real effective exchange rate is $d\log q_A^E \equiv \frac{1}{N-1}\sum_{j\neq A} d\log q_{Aj}$. With P_{Ni} fixed, $d\log P_i = m\, d\log P_{Mi}$, so $d\log q_{Aj} = d\nu_j + m\,(d\log P_{Mj} - d\log P_{MA})$. Let A impose t and its partners impose none. With equal import weights, $d\log P_{MA} = d\nu_A^E + \bar{t}$ and $d\log P_{Mj} = d\nu_A^E - \frac{N}{N-1} d\nu_j$, so

$$d\log q_{Aj} = \Big(1 - \frac{mN}{N-1}\Big) d\nu_j - m\bar{t}, \tag{C.1}$$

where $-m\bar{t}$ is the direct effect of the tariff on A's consumer price index. For a unilateral tariff, $\bar{t} = d\tau/(N-1)$, and averaging with Eq. (3.8),

$$\frac{d\log q_A^E}{d\tau} = -\Big(1 - \frac{mN}{N-1}\Big)\frac{\rho_D}{(N-1)M_N} - \frac{m}{N-1}. \tag{C.2}$$

Proposition C.1 (Real exchange rates)**.** *Consider a marginal unilateral tariff $d\tau > 0$ by A on B at the symmetric benchmark, $N \geq 3$, $m > 0$, partners imposing no tariffs, and define*

$$H_N(\rho) \equiv \Big(1 - \frac{mN}{N-1}\Big)(\rho - N\rho_D) - mN\, M_N(\rho).$$

For every untariffed bystander c, if $M_N(\rho) \neq 0$, $d\log q_{Ac}/d\tau = H_N(\rho)/[N(N-1)M_N(\rho)]$, so wherever MML holds the real bilateral response has the sign of $H_N(\rho)$. Since ρ_D does not depend on ρ, $dH_N/d\rho = 1 - mN$.

(i) If $m < 1/N$, H_N has the unique root

$$\rho_q^*(N) = \frac{[(N-1) - mN]N\rho_D + mN[N\alpha_D(\eta-1)+1]}{(N-1)(1-mN)}.$$

Where MML holds, A appreciates in real terms against c for $\rho < \rho_q^*(N)$ and depreciates for $\rho > \rho_q^*(N)$; if MML holds at $\rho_q^*(N)$, then $\rho_q^*(N) > N\rho_D$.

(ii) If $m = 1/N$, H_N does not depend on ρ: $H_N = -[1 + N\alpha_D(\eta-1) + N(N-2)\rho_D]/(N-1)$.

(iii) If $m > 1/N$, H_N is decreasing in ρ, so any crossing has the reverse orientation.

(iv) If $\eta \geq 1$, so that MML holds for every $\rho > 0$, and $m \geq 1/N$, then $H_N(\rho) < 0$ for every $\rho > 0$: no positive ρ produces a real depreciation against an untariffed bystander.

(v) Under MML, if $m \leq (N-1)/N$, A's REER appreciates. Under weak home bias this condition holds automatically, since $m = \alpha_T(1-\alpha_D) \leq (N-1)/N$.

At the US-like calibration and $N = 3$, $m = 0.08 < 1/3$ and $\rho_q^* = 4.93$, against the nominal threshold $3\rho_D = 3.96$.

Proof of Proposition C.1. Let $\bar{t} = d\tau/(N-1)$. Substituting Eq. (3.8) into Eq. (C.1) gives the stated response $H_N/[N(N-1)M_N]$, which has the sign of H_N under MML. Since M_N has slope $(N-2)/(N-1)$ in ρ, $dH_N/d\rho = 1 - \frac{mN}{N-1} - \frac{mN(N-2)}{N-1}$, which equals $1 - mN$.

(i) Solving $(N-1)H_N = 0$, the coefficient on ρ is $(N-1)(1-mN)$, which yields $\rho_q^*(N)$. At the root $[(N-1) - mN](\rho_q^* - N\rho_D) = mN(N-1)M_N(\rho_q^*)$, which is positive if MML holds there; since $(N-1) - mN > N - 2 > 0$ for $m < 1/N$, $\rho_q^*(N) > N\rho_D$. (ii) and (iii) follow from the sign of the slope, with the constant in (ii) obtained by substituting $m = 1/N$ into H_N.

(iv) If $\eta \geq 1$, every term of M_N is nonnegative and the first is positive, so MML holds for every $\rho > 0$, and $\rho_D - m \geq 1 - 2m$ implies $H_N(0) < 0$ (online supplement); for $m \geq 1/N$, $H_N(\rho) \leq H_N(0) < 0$.

(v) In Eq. (C.2) the first term is nonpositive if and only if $m \leq (N-1)/N$, and the second is strictly negative. Under weak home bias, $m = \alpha_T(1-\alpha_D) \leq 1 - \alpha_D \leq (N-1)/N$. □

C.3 Data notes for the 2025 episode

Exchange-rate data. Bilateral rates and the nominal broad dollar index (goods and services) are from the Federal Reserve H.10 release. Rates are expressed in dollars per unit of foreign currency (the CNY, CAD and MXN quotes are inverted), and the broad index is inverted, so a rise is a dollar depreciation. The index excluding China removes the renminbi from the daily log change of the broad index using China's 2025 H.10 weight (10.9 percent) and divides by one minus that weight. No daily dong series is available; the VND row of Table 1 uses BIS monthly averages in the monthly columns and BIS end-of-month rates in the daily columns (marked [a]). "Monthly" compares monthly averages, January with March (Phase 1), March with April (Phase 2) and July with December (Phase 3); "daily" runs from January 31 to April 2, April 2 to May 12 and August 6 to December 31. Panel (a) of Fig. 5 plots monthly averages relative to each series' January 2025 average; the August dong average is missing in the source.

Tariff data. Panel (b) of Fig. 5 plots statutory headline tariffs above the pre-2025 baseline by effective date: US tariffs on China, China's across-the-board tariffs on the United States, US tariffs on the European Union, the 10 percent US baseline on unlisted partners, US tariffs on Canada, and Canada's tariffs on covered US goods. Rates are statutory headline rates, not trade-weighted measures. Sources are Federal Register executive orders, the Department of Finance Canada, and USDA Foreign Agricultural Service summaries of State Council Tariff Commission announcements. The complete chronology is in the online supplement.

Phase definitions. Phase 1 (February 1–April 2) covers the two US tariff increases on China and China's product-specific responses. Phase 2 (April 3–May 12) covers the US–China escalation, the 10 percent baseline of April 5, from which Canada and Mexico were excluded, and the higher

country-specific rates of April 9, which were suspended for every partner except China on April 10. Phase 3 (August 7–December 31) begins with the August framework rates.

Classification caveats. The phase classifications are approximate. Tariffs on Canada and Mexico, partial retaliation, exclusions, and contemporaneous policy announcements mean that none of the phases is a clean experiment. Vietnam also shifted between bystander-like and targeted treatment during the year. The online supplement documents these complications, the complete tariff chronology, sector-specific factors, and permanent trade imbalances.